



POLITECNICO
MILANO 1863

SOFTWARE ENGINEERING II
COMPUTER SCIENCE AND ENGINEERING

Requirement Analysis and Specification Document

Students&Companies

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Academic Year: 2024-2025

Deliverable:	RASD
Title:	Requirement Analysis and Verification Document
Authors:	Nardi Andrea Pesaturo Christian Giovanni Pinessi Andrea
Version:	1.0
Date:	22-December-2024
Download page:	GitHub
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1 Introduction

1.1 Purpose

The **Students&Companies** project, as its name suggests, aims to bring university students closer to the corporate world. The system enables the creation of a network linking students to companies through a dedicated portal, facilitating the meeting between the two realities within an application context. Member universities can monitor the status of activities and interactions between students and companies, ensuring effective supervision of the relationships established.

1.1.1 Goals

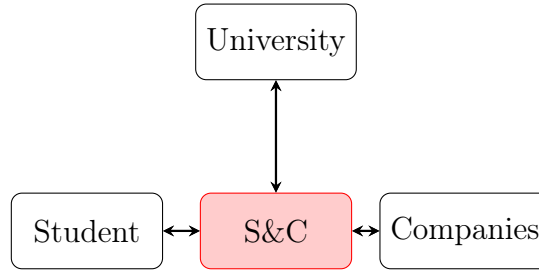
- **G.1:** Facilitate the connections between students, universities and companies for internships.
 - **G.1.1:** Connect university students seeking internships with companies offering them.
 - **G.1.2:** Companies can search for students based on their skills, experiences, and aptitudes, as well as the specific needs of the companies.
- **G.2:** Improve the quality of applications and offers.
 - **G.2.1:** Provide personalized suggestions to students to optimize their CVs.
 - **G.2.2:** Offer guidance to companies to enhance the descriptions of their projects, increasing the chances of a successful match.
- **G.3:** Effectively manage the selection process by supporting companies in organizing interviews and collecting information from candidates, facilitating decision-making.
- **G.4:** Monitor the execution of internships by enabling universities to oversee their progress, ensuring that students receive a proper educational experience and that companies adhere to the agreed terms.
- **G.5:** Promote collaboration between students, companies, and universities by creating an integrated ecosystem that encourages communication and cooperation among all stakeholders (students, companies, universities) involved in the internship process.

- **G.6:** Improve the quality of matches by using advanced recommendation mechanisms, from keyword matching to statistical analysis, to achieve superior pairings.
- **G.7:** Support stakeholders in problem resolution by providing mechanisms that allow students, companies, and universities to report, address complaints, and terminate the internship.

1.2 Scope

The aim of the Students&Companies platform is to channel the interaction and identification between university students and companies for internships and apprenticeships. Through statistics on the basis of students' CVs and their profiles and what companies offer. S&C also considering mapping with all interactions between the two entities.

Phenomena This section aims to describe and illustrate what are the connections and dialogues between the *student* and the *company* through the platform, conveying information also towards a third entity: the *university* that has the task of monitoring the nature and quality of the relationship between the *student* and the *company*. We will see the phenomena that enable the interaction between the different entities.



1.2.1 World Phenomena

Word Phenomena concern events of the real-world domain in which the users can interact with each other and with the environment but not directly with the system.

<i>ID</i>	<i>Description</i>
W01	Students seek internship opportunities.
W02	Students start the internship.
W03	Students end the internship.

<i>ID</i>	<i>Description</i>
W04	Companies identify the need for new trainees for their projects.
W05	Universities establish guidelines and policies for student internships.
W06	Changes in business needs influence the type of internships offered.
W07	Reasons outside the application cause changes in the actual placement.
W08	Companies look for ways to make their offers more attractive to the best students.
W09	Students seek advice on how best to present their skills and experience.
W10	University staff intervene to solve problems reported in company internship policies (not interaction with the system).

1.2.2 Shared Phenomena

Shared Phenomenon refers to events or facts that can be observed by both the system and its surrounding environment. This allows the system to influence the environment while also being influenced by it.

<i>ID</i>	<i>Description</i>
S01	Students sign up to the platform.
S02	Students decline an internship opportunity.
S03	Companies register on the platform.
S04	Companies publish and update internship offers via the system.
S05	Students use the system to search for and apply for internships.
S06	University staff receive complaints or problems reported through the system.
S07	Students receive suggestions on how to improve their CVs.
S08	Companies receive suggestions from the system on how to improve internship offers.
S09	Students report problems or complaints through the system.
S10	Companies interested in internships report complaints or grievances via the system.

<i>ID</i>	<i>Description</i>
S11	Companies update the apprenticeship offers in the system according to changes in their needs.
S12	Students update their profiles and CVs in the system.
S13	Students and companies communicate via the platform.
S14	Universities monitor the intership situation through the application.

Machine Phenomena are events that occur automatically thanks to the system's algorithms and logic. These events happen independently of the direct action of users and are essential for creating interactions between the parties. They play a crucial role in the functioning of the platform.

<i>ID</i>	<i>Description</i>
M01	The system processes the matching of students and placements, creating communication between the student and the selected company.
M02	The system generates suggestions for students and companies.
M03	The system collects feedback and statistics.
M04	The system manages data storage and retrieval (Storage: "Student profiles", "Internship offers from companies", "Feedback and suggestions", "Complaints and reports", "Interaction history", "Analytical data". Data retrieval: "Display information to users", "Generate customised suggestions", "Monitor and manage processes").
M05	The system notifies users of new traineeship opportunities or messages.
M06	The system allows monitoring of traineeships by universities.
M07	The system handles reports of problems and complaints.

1.3 Definitions, Acronyms, Abbreviations

Recommendation:

- The **Student** receives a notification regarding an internship potentially of interest to them.
- The **Company** receives a notification regarding one or more compatible **Students**.

Important: The recommendation does not constitute an application. It only becomes an application when the student reviews the recommendation and decides to accept it.

- The company can express its interest in a student through positive feedback during the application evaluation phase.

Acronyms

- I. S&C: Students&Companies
- II. UML: Unified Modeling Language
- III. CV: Curriculum Vitae

1.4 Revision History

Version	Date	Changes
0.1	11/19/2024	Goals, Phenomena, Requirements, Scenarios, Initial Drafts and Document Structure.
0.2	11/25/2024	State Diagram, UML and User Characteristics.
0.3	11/27/2024	Assumption, Dependencies, Constraint, and External Interface Requirement.
0.4	12/02/2024	Functional Requirement.
0.5	12/03/2024	Use Cases + Starts Sequence Diagram.
0.6	12/06/2024	End Use Cases.
0.7	12/06/2024	End Sequence Diagram.
0.8	12/08/2024	Start Alloy + GUI.
0.9	12/14/2024	End Alloy.

Version	Date	Changes
1.0	12/17/2024	Final Recap.

1.5 Reference Documents

The specification document Assignment RDD AY 2024-2025.pdf

1.6 Document Structure

Chapter 1: Introduction This chapter introduces the project, explaining its purpose, objectives, and intended audience. It outlines the context in which the software will be used, providing an overview of relevant information to understand the application domain. Additionally, it clarifies the main goals of the project, along with a list of related phenomena and objectives. Furthermore, it includes a table summarizing the team's effort and highlights the key meetings that shaped and guided the progress of the project.

Chapter 2: Overall Description This chapter describes the main entities of the system through a domain-level class diagram, which is instrumental in understanding the conceptual structure of the software. It also includes state diagrams that define the execution flow of actions. Scenarios are provided to illustrate how the system behaves in typical situations. The main functionalities, characteristics of the target users, and assumptions made are discussed. Additionally, the operational context of the software is analyzed, emphasizing its dependencies, constraints, and relationships with other systems.

Chapter 3: Specific Requirements This chapter focuses on the technical details of the system. It explains how the system connects and interacts with external actors through interfaces. Requirements are listed in a clear and structured manner. To enhance comprehension, use cases and their associated sequence diagrams are included to better explain the system's behavior.

Chapter 4: Formal Analysis with Alloy This chapter is dedicated to the formal analysis of the system using Alloy. It presents the Alloy code written to validate the system's properties, detailing the decisions made and the techniques employed. The fundamental structures of the system are outlined, followed by facts, predicates, functions, and assertions that ensure the model's consistency.

Chapter 5: Effort Summary This final chapter provides a summary of the work carried out, specifying the time allocated to various project activities. A comprehensive estimate of the effort required to complete all development phases is also included.

2 Overall Description

This section provides an overview of the system, including high-level functionality and design constraints. Flow diagrams and UML diagrams will be presented and illustrated to explain how the system works, adopting a pedantically technical approach, albeit sufficiently precise and comprehensive.

2.1 Product Perspective

To enable a better understanding of the logical progression and various states within the S&C platform, we aim to provide a logical progression represented through a **State Diagram** and **UML Class Diagram**. The representation is high-level but encompasses the entire dynamics of the platform.

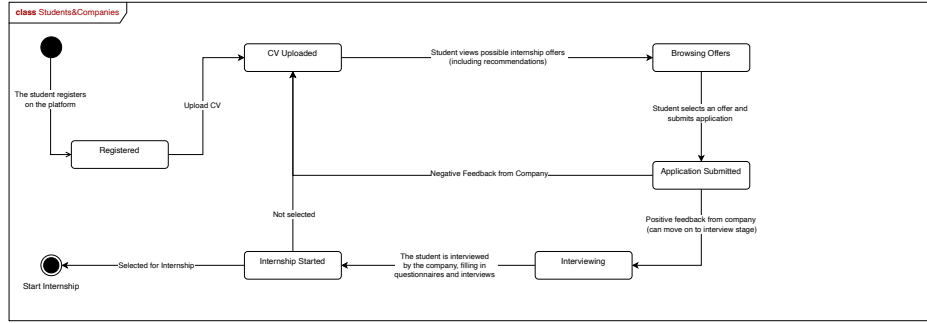


Figure 1: State Diagram Student

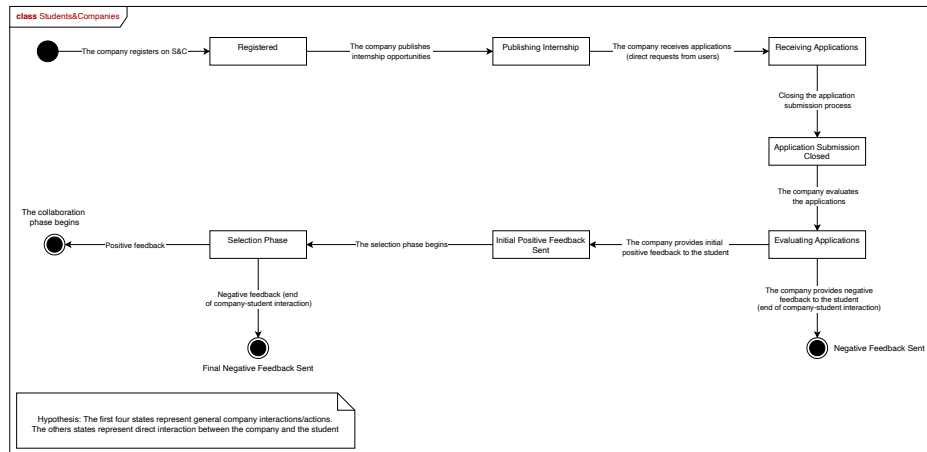


Figure 2: State Diagram Company

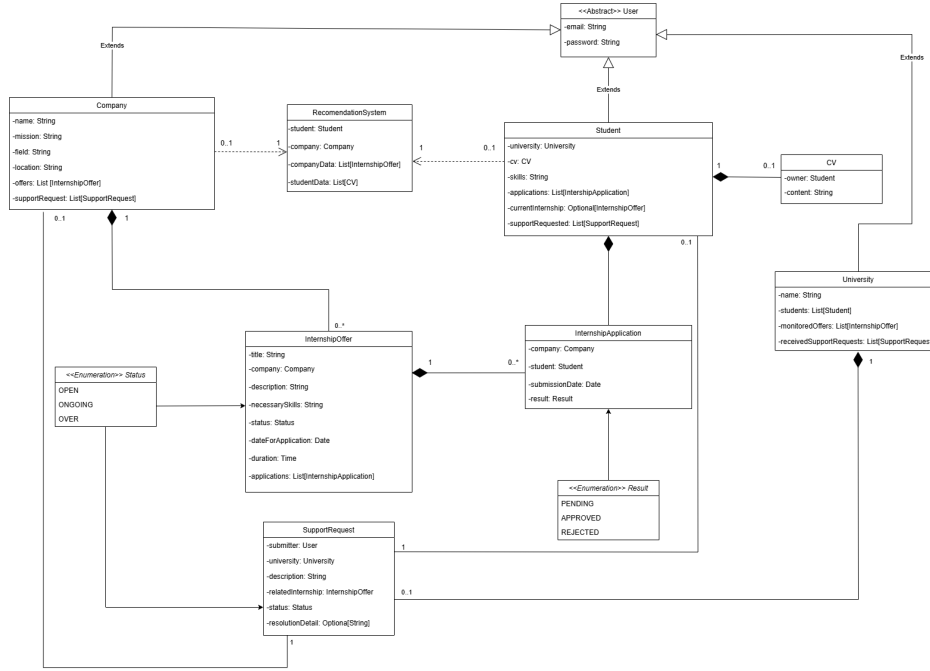


Figure 3: UML Class Diagram

We deemed it necessary to divide the state diagrams for the two actors: Students and Companies, to make the diagram structure more understandable and focused. When referring to the evaluation of applications, it includes both direct applications submitted proactively by students and those received through the system’s automated matching process (which the company may choose to accept or decline).

2.1.1 Scenarios

S1: The student registers on the platform (entering preferences for the type of internship and desired position)

Student M, a university student seeking an internship opportunity, decides to register on the Students&Companies (S&C) platform to explore available offers and receive personalized recommendations. M opens the browser and searches for the platform using a search engine. After finding the official S&C website, he selects the “Student Section” page from the menu and then presses the “Register as Student” button. The system redirects M to the registration page, where he is guided to enter the required personal information. During the registration process, M completes the following mandatory fields: First and Last Name, University Email Address (for

account creation and future communications), Password (to create and confirm), Year of Birth, Affiliated University, Degree Course. Subsequently, the system asks M to specify some initial preferences to personalize the experience on the platform. These preferences include: Preferred type of internship (e.g., curricular, extracurricular, paid/unpaid). Desired geographical location (M selects “Milan”). Maximum or minimum duration of the internship. On-site or remote. After completing all the required fields, M presses the “Register” button. The system verifies the correctness of the entered data and sends a confirmation email to the specified address. M receives the email and clicks on the verification link to activate his account. Once verified, the system confirms the creation of the account with a welcome message and redirects M to his personal area. From here, the student can upload the CV and start searching for personalized internship offers.

S2: Creation and Improvement of the Student’s CV

To start searching for offers, the student must upload their CV if they haven’t done so previously. They access the “Curriculum Vitae” section in their profile within the application. Here, they click on the “Add a new CV” button and upload their resume in PDF format, previously created and saved on their computer.

Once uploaded, they receive a pop-up message confirming the successful completion of the procedure and displaying the newly added PDF.

At a later stage, the student may decide to make changes to their CV, also thanks to the advice provided by the automated analysis system. To update the CV, they access the “Curriculum Vitae” section in their profile again. Next to the previously uploaded CV, they find the “Edit” button; by clicking it, they can select a new PDF file with the updated CV.

As before, after uploading, they receive a pop-up confirmation message and can view the newly uploaded PDF.

S3: Company Registration

Company X, interested in participating in a collaboration program for internship positions with university students, registers on the S&C platform. It accesses the website through a search for the platform on a search engine, selects the “Company Section” page from the menu, and then presses the “Register as Company” button. The registration phase requires entering information such as “Legal Name of the Company,” “VAT Number,” “Address

of the Main Office,” “Sector of Activity,” “Company Mission Description,” etc., and also a contact person appointed by the company, including “Email Address,” “First and Last Name,” “Job Position,” and “Phone Number.” After receiving and evaluating the company’s registration request on the S&C portal, a staff member will send the outcome of the evaluation to the company’s contact person, using the phone number or, alternatively, the provided email address.

S4: Posting Internship Positions

Company Y, interested in offering internship opportunities, accesses the Students&Companies (S&C) platform using its corporate account. After logging in, Y selects the “Post Internship Offer” option from the main menu. The system redirects Y to a dedicated form, where it is necessary to enter the following information:

Internship title, internship description, expected duration, internship location, type of internship (curricular/extracurricular).

The company must also set a deadline for the application phase, which defines the final date for students to submit their applications. After this date, the system will automatically close the possibility to apply and will activate the selection phase.

After completing all the sections, Y confirms the publication of the offer. The system verifies the data and shows a preview of the announcement, requesting a further confirmation. Once approved, the offer is published and made visible to students on the platform.

Y receives a confirmation notification of the publication and can monitor in real-time the number of applications received, preparing for the next selection phase once the deadline has passed.

S5: Recommendation and Matching System

Once both students and companies have completed their profiles, the S&C system comes into action. Using artificial intelligence and machine learning algorithms, it analyses the provided data to identify the best matches between students and internship offers. The system considers various factors, such as skills compatibility comparing those required by companies with those possessed by students the alignment of professional interests, and compatibility in terms of time availability and logistical preferences. The system generates for each student a personalized list of internship

offers, ordered based on the degree of compatibility, and notifies the student whenever this list is automatically updated. Each offer is accompanied by a brief description and the reasons why it was suggested, helping the student understand the alignment with their own profile. Students receive notifications about new relevant offers as soon as they are published, ensuring they do not miss out on important opportunities. Students can explore the suggested offers, save those of interest, and apply directly through the platform. They have the option to filter and sort the offers based on additional personal criteria. Student recommendations become applications only when the student accepts an automated recommendation received. Therefore, companies can view the automated recommendations in the list of all received applications (once they have been accepted by the respective students). They will be visible within a specific subsection of recommendations defined as “Received Application - Recommended”. The system encourages users to provide feedback on the recommendations received. Students can report if an offer is not relevant or particularly interesting, and this feedback is used to improve future personalized recommendations. Companies can assess the relevance of the suggested profiles, helping the system better understand their specific needs.

S6: Student’s Spontaneous Application for an Available Internship from the Internship List

Student A accesses the S&C platform; after the required login phase, he selects the “Internship and Collaboration” option in the menu, where he views a list of currently available and open internships. Once he identifies the position of his interest, he clicks on “Apply”; he is then asked to fill out a brief cover letter. After reviewing what he has written, the student clicks “Submit Application” and will receive a notification and an email confirming the submission of the application. Subsequently, the company will receive the application and, once the period for collecting applications has ended, can start the selection process.

S7: Internship Withdrawal and Application Cancellation Process

Student A accesses the S&C platform, navigates to the “Internship and Collaboration” section, where they can view a list of currently available and open internships, including those they have applied for.

The student can withdraw their application using the “Withdraw Ap-

plication” button, upon which they will receive an email confirming the “Application Successfully Withdrawn.” Company X will receive a “Application Withdrawn” notification for Student A only if the student is involved in the stages following the “Application” phase, i.e., steps occurring after the internship application deadline.

S8: Management of the Selection Process by the Company

Once the application period for students has ended, Company A begins the selection phase. The company manager logs into the system with their credentials and goes to the “Received Applications” section. Within this section, they find all the applications received for that specific internship.

For each application, the system shows the manager a series of key data such as: the student’s first and last name, affiliated university, course of study, the student’s CV, and a compatibility score. This score is automatically calculated by the system based on certain automated evaluations. It helps the company to identify a shortlist of candidates who best match the desired characteristics. In this way, the company can immediately discard a good number of candidates who do not fully meet the sought-after criteria. All students eliminated in this initial phase will be included in the group of non-selected students, who will receive a ‘Participation Acknowledgments’ message. Once the pool of candidates has been narrowed down, the manager sends each selected candidate one or more structured questionnaires to assist in making the final decision. Students receive an email and an internal system notification informing them of the need to complete the questionnaires in order to proceed with the selection process.

Candidates access the platform and find the assigned questionnaires in their personal area. Each questionnaire may include multiple-choice questions, true or false, open-ended responses, or practical exercises.

Once the candidates have submitted their responses, the system proceeds with the automatic evaluation of the questionnaires.

The manager receives a notification and an email when the results are ready. By accessing the list of all applications, they can now view a detailed report for each candidate, which includes: a total score obtained in the questionnaires, a skills analysis highlighting strengths and areas for improvement, and a comparison with the average of other candidates.

With this information, the manager can make an objective assessment of the candidates without the need for direct interviews. If necessary, they can also consult the curriculum vitae to have a more complete picture of each profile. Once this is done, they will draw up a ranking among all the remaining

candidates.

At this point, the manager will send each chosen candidate an internship offer, using the appropriate “Accept Application” button found within each application. This will include, as a summary, the data previously seen by the student and new information, such as: start and end date of the internship, daily working hours, daily entry and exit times from the company, description of expected activities and responsibilities, any compensation and benefits, and information on how to accept the offer.

To the non-selected students (including those initially discarded), once all the selected students have accepted the offer, the manager will send a "Participation Acknowledgments" for participation using the appropriate “Reject Application” button.

In the event that a student declines the offer proposed after their selection, the manager will be notified within the platform and by email, and can decide whether to select another student from those subsequently present in the previously drawn-up ranking.

S9: The student has been selected for an internship in a company

The student L receives the offer in their personal area and via an email notification. They access the “Offer Management” section and can view the complete history of received offers. Among these, they will find the new offer. By clicking on it, they can view all the details and either accept or reject it directly through the platform. By clicking the “Accept” button, the offer will be accepted, and positive feedback will also be sent to the company in question. At this point, the system updates the application status to “Internship Accepted,” preventing the student from starting any new internships until the conclusion of the one just accepted. If the student decides to reject the offer, prompt feedback will be sent to the company, which can then decide whether to extend the offer to the next candidate in the ranking established during the selection phase.

S10: Monitoring of internships by the university

University P, responsible for monitoring the internships carried out by its students, accesses the S&C platform using an institutional account. After logging-in, a university representative selects the “Internship Monitoring” option from the main menu. The system displays a dashboard with a detailed list of all internships associated with the students of University P. For each

internship, key information is displayed, such as: student name, hosting company, internship status (e.g., ongoing, completed, interrupted).

The university representative can select a specific internship to obtain further details, such as the activity log, any reported issues, start and end dates of the internship, and daily working hours.

S11: Management of reports and complaints by the university

University U, after logging into the S&C platform, notices a notification related to the internship of Student A. Notifications are color-coded to indicate their status: red for new unread reports; yellow for ongoing but unresolved issues; and green for internships without any problems. By clicking on the notification, the university representative accesses the “Complaint Management” section, where they can read the report submitted by the student. Using the communication tools integrated into the platform, the university contacts the company representative to request clarifications or additional information. Once the issue is resolved, the university representative updates the complaint status to “resolved,” automatically notifying Student A via email. The process is similar if Company X submits a report concerning Student G. In this case, the university representative uses the platform to contact the student directly, gathering details or proposing solutions to address the problem. In both cases, the university can request further clarifications from the party that originated the report before proceeding with a resolution. This ensures effective and traceable problem management, improving communication among all parties involved. Student A and Company X will have a shared section, also accessible to University U, under the “Activity Monitoring” tab. This section will be visible and editable by both the company and the student, enabling them to provide updates to University U on the progress of the collaboration agreement. In the event of severe issues in the Student A – Company X relationship, University U may request the early termination of the collaboration agreement.

S12: Suggestions and CV Improvements (Feedback for Students and Companies)

Student X, registered on the S&C platform, wants to improve their CV to increase the chances of securing an internship. The student logs into their S&C personal area and then accesses in the "CV Suggestions". The sys-

tem automatically analyzes X's CV using recommendation mechanisms that compare the skills and experiences listed with the descriptions of internship offers available on the platform. Based on this analysis, the system provides personalized suggestions, such as: "Add details about the technologies used in your previous projects, such as Python or SQL, to align with the demands of companies active in the IT sector." "Include an initial summary highlighting your most relevant skills, such as problem-solving abilities or teamwork experience."

Similarly, Company Y, which posts an internship offer on the platform, can access the "Offer Management" section. Here, the system evaluates the project description and offer terms, providing suggestions to enhance the appeal of the posting. For example: "Clearly indicate the benefits offered, such as specific training or mentorship opportunities, to increase applications." "Specify the skill level required for technologies like Java or React, avoiding generalizations."

These suggestions help students and companies optimize their CVs and internship postings, respectively, improving the recommendation process and increasing the likelihood of finding the ideal match. The system also allows saving suggestions for future updates, ensuring continuous support in the professional development of students and the enhancement of company offers.

S13: Feedback and Statistics Collection

The S&C system automatically collects feedback from Student A, Company X, and University Z regarding internships and interactions on the platform through a dedicated section called "Feedback and Improvements" accessible via the menu. Users can provide ratings on aspects such as "Overall Satisfaction," "Communication and Support," and "Matching Adequacy." Specifically, Student A can evaluate "Clarity of Internship Offers" and "Ease of Application," while Company X can assess "Support During the Selection Process" and "Alignment of Student Profile with Required Profession."

This data is processed to generate statistics useful for analysis, feeding and enhancing the system and its recommendation process.

S14: Edit Student Profile to Change Internship Preferences

Wishing to modify internship preferences, Student C accesses their profile in the application and navigates to the "General Data" section. Among the available options, they find "Internship Preferences." Within this section,

they can update their preferences regarding the type of internship sought and its maximum duration. After making the desired changes, they click the “Save” button at the bottom of the page, ensuring that their new preferences are correctly saved.

2.2 Product Functions

Here we include the most important requirements for the product.

User Registration and Login

The S&C platform must enable students to register and create a detailed profile containing personal information, field of study, ambitions, past experiences, and acquired skills. Companies must be able to register their profiles, post internships, monitor received applications, and communicate directly with students. The platform will allow universities to access data related to student internships, monitor internship activities, and manage any complaints that may arise between students and companies.

A secure login system and credential management must be implemented. During the registration and login process, users will select their appropriate role (e.g., student, company, or university).

It will not be possible for a registered user to simultaneously hold both a student and a company role, as this would exceed the primary objectives of the software. The system will prevent a student from accessing functionalities intended for companies or universities, and vice versa.

Posting an Internship Offer

This feature will be available to registered companies. Each company will have the ability to post internship offers by logging into its account on the S&C platform and selecting the “Post Internship Offer” option. The system will guide the company through a dedicated form where they can input essential details, such as: Internship title, a detailed description, expected duration, location, type of internship (curricular or extracurricular), required skills, deadline for application submission.

After completing the form, the system will display a preview of the announcement, allowing the company to review the entered details. Once approved, the announcement will be published on the platform and made visible to students.

The company will receive a confirmation notification upon publication and will gain access to a dashboard to monitor the number of applications received and view candidate information. At the application deadline, the system will automatically close the applications and initiate the next selection phase.

A Student's Application for an Internship

This feature allows students to apply for internships through the S&C platform. After companies publish internship opportunities, students can browse the job board and select the internship of interest. They must complete all required fields in the application form. Once submitted, the student will receive a confirmation email and wait for feedback from the company. The student will be able to withdraw their application at any point during the internship application and selection process. If the student withdraws their application before the application phase closes, the company will not be notified. Conversely, during later stages of the process, the company will receive a notification informing them of the student's withdrawal.

Complaints Management and internship Status

Students, Companies, and Universities can access a dedicated section to monitor the progress of an internship after it has started. Only Universities will have the ability to view complaints between Students and Companies.

A complaint submitted by a Student against the host Company will not be immediately visible to the Company. The University will act as a mediator to address and resolve any issues that arise. In critical situations, the university may request the early termination of the internship. The student and the company will provide updates to the university to enable monitoring of the internship's progress.

Feedback and Statistics Collection

The system will collect feedback and statistics from Students, Companies, and Universities. In the dedicated section, users can provide input to the system through structured question forms or open-ended responses, highlighting areas where improvements and adjustments are needed.

The collected data will be used to refine the platform and improve the recommendation process.

Internship Monitoring

This feature will be available to Universities registered on the platform. Each university will have the ability to monitor the internships undertaken by their students. After logging in, the university representative will access a centralized dashboard with a detailed list of all active and completed internships involving their students.

For each internship, key information will be displayed, including: student's name, host company, status (e.g., ongoing, completed, interrupted), feedback from both the student and the company.

The university representative can delve deeper into each internship to view additional details, such as: activity logs, reported issues, start and end dates of the internship.

In case of reported issues, the university representative can intervene using the "Manage Complaints" option. This feature allows direct communication with the company via the platform or the arrangement of a meeting to discuss and resolve the situation.

The platform will also provide the ability to log interventions and corrective actions taken, ensuring timely and accurate management of internship-related issues.

CV Suggestions for Students

This feature will be available to students registered on the platform. Each student will have the opportunity to enhance their CV through the "CV Suggestions" section accessible from the platform's main menu. Once logged in, students can upload their CV or update it directly on the platform. The system will perform an automated analysis of the CV, comparing it with the published internship opportunities. Based on this analysis, the system will provide personalized suggestions to improve the CV. These suggestions may include, for example, adding specific skills, improving formatting, or including relevant work experiences. Students will have the ability to view, accept, and implement the suggestions provided, updating their CV accordingly. Additionally, the system will allow students to save suggestions for future updates, ensuring continuous improvement of their professional profile and increasing their chances of securing an internship.

Internship Offer Suggestions for Companies

This feature will be available to companies registered on the platform. Each company will have the opportunity to optimize its internship offers through the "Offer Management" section accessible from the platform's main menu. After logging in and publishing a new offer, the system will automatically analyze the internship description and provide suggestions to enhance its appeal. These suggestions could include adding details about offered benefits, specifying the required skills more precisely, or restructuring the text to make it more engaging for students. Companies will be able to review the suggestions and decide whether to apply them to improve the quality and clarity of their offers.

Recommendation and Auto-Matching

The system must provide an automated recommendation feature for both students and companies.

This functionality should allow students to be notified when a new internship matching their characteristics becomes available and notify companies when a new CV fitting their needs is uploaded.

The automated recommendation process is a bilateral procedure: in fact, when an internship is recommended to a student, the CV of the same student will also be recommended to the company. Thus, the recommendation of a student to a company will become an "Actual Application" only when the student decides to accept the system's recommendation for the internship at that company.

The company will then be able to view all Recommended Applications grouped and highlighted. It will still be possible to see all other spontaneous applications submitted by students.

Selection Assistance

The selection assistance functionality will be available for companies offering internships.

This feature will guide companies through the selection process by providing specific, well-formatted questionnaires to send to students.

These questionnaires will enable the effective execution of the selection process. They will include multiple-choice questions, true or false, open-ended questions, or practical exercises. The system will ensure that questions are

presented clearly and that sufficient time is allocated for their completion. For closed-ended questions, the score will be calculated immediately. For open-ended questions or practical exercises, the system will provide tools for manual evaluation by the responsible examiner (such as scoring grids and evaluation criteria).

Once both types of scores are finalized, a ranking will be generated by the company examiner on the platform. The ranking will display all candidates in order from the best to the least suitable, facilitating the sending of “Internship Offers” to the selected candidates and “Participation Acknowledgments” to those not selected.

Notification

This functionality will be essential for many processes managed by the platform.

Notifications will frequently need to be sent to companies or students. The notification process will ensure that for every notification, the relevant user will be informed via email (to the institutional email address used for registration on the platform) as well as through the website.

Each user will, therefore, be able to find their notifications either in their email inbox or within directly on the platform.

Some important examples include: initial activation of a student’s profile, registration confirmation for a company, confirmation of an application submission by a student, receipt of an automated recommendation, receipt of an internship offer, rejection of an internship offer by a student sent by a company.

Student Profile Management

This functionality will be available exclusively for students.

It will allow students to modify their preferences over time regarding various aspects, such as the duration of internships, desired internship characteristics, uploading a CV, and updating it. This functionality will be available only to students registered on the platform.

Students will have the ability to upload their Curriculum Vitae (CV) to the platform, which will then be reviewed and used during the analysis and selection phase by companies offering internships.

Searching for a new internship will only be possible if the CV has been previously uploaded.

Once the first CV has been uploaded, students will have the option to update it over time by uploading new, updated versions.
The accepted file format for CV uploads is PDF.

Application Compatibility Calculation

This functionality will be available to companies.
It is designed to assist companies in initially filtering out candidates who do not precisely match their requirements by automatically calculating a compatibility score between students and the company.
The feature uses specific algorithms to analyze student characteristics, including a detailed assessment of their CVs.
Based on this data, the system will generate a specific compatibility score for each student relative to the company.
The score will be displayed among the student data for each received application.

2.3 User Characteristics

In this section the main characteristics of the users, directly involved with the system, are outlined.

2.3.1 Student

Students have the opportunity to register on the portal to search for and take advantage of internship offers available from companies within the system. Through the portal, a student is able to: modify their CV, interact with companies, submit their application for an open position, communicate the progress of the internship, modify their internship preferences and make possible complaints.

2.3.2 Company

By registering on the S&C platform, the **Company** can offer various types of contracts for collaborations with university students. The company creates the announcement and outlines the open positions along with the available spots. It then carries out the selection phase and subsequently assigns the collaboration tasks to the selected students. In case of issues in the relationship with the student(s), the company can submit complaints on the platform.

2.3.3 University

The **University** offers students the opportunity, through registration on the platform, to establish collaborative relationships of various kinds with companies participating in the initiative. The university constantly monitors student progress and internships at companies and has the ability to view and manage complaints shared on the portal.

2.4 Assumptions, Dependencies, and Constraints

In this section, we will present assumptions that, throughout the document, will be considered indisputably true. Therefore, it is important to clearly define these assumptions within this specific section. No scenarios in which the following listed assumptions are found to be false will be considered. Additionally, the constraints and dependencies influencing the system will be described to ensure a comprehensive understanding of the limitations and external relationships that may impact development.

<i>ID</i>	<i>Description</i>
A01	The university is already registered on the platform through an initial phase where it fills out a contract and establishes a formal agreement with S&C.
A02	It is assumed that both students and companies provide accurate, complete, and up-to-date informations at all times. This ensures the proper functioning of the system’s matching features.
A03	All students possess a university email address, as this is required for signing up and verifying their identity on the platform.
A04	Students are interested in participating in internships.
A05	Companies are capable of offering internships, paid placements, and/or providing both tangible and intangible benefits, such as training and mentorship.
A06	It is assumed that students and companies are interested in receiving suggestions to improve their CVs and internship offers, and that they will use these suggestions to enhance their chances of success.
A07	It is assumed that most interactions between students, companies, and universities will occur through the S&C platform, ensuring traceability and ease of management.
D01	The system depends on external services to send emails to communicate with users in a timely manner.
C01	The system must be accessible from various devices (PC, tablet, smartphone) and compatible with major web browsers.

<i>ID</i>	<i>Description</i>
C02	Students register using their institutional email address.

3 Specific Requirements

Here, detailed functional and non-functional requirements of the system are specified.

3.1 External Interface Requirements

Here we describe the various external interfaces required for the product.

3.1.1 User Interfaces

Users: **Students**, **Companies** and **Universities** will interface with **S&C** Platform via Website. The website interface will be designed as a set of structured pages that will provide users with the necessary functionality to interact with the platform in an intuitive and efficient manner.

Web Pages Overview

Registration Page: This page enables students and companies not yet registered on the **S&C** platform to select the appropriate pathway, directing them to the respective registration form.

Home Page for Login or Registration: Displays key information about the platform's mission and services. The menu offers options to select "*Student Section*", "*Company Section*" and "*University Section*".

Student Section: Provides the option "*Register as Student*".

Company Section: Provides the option "*Register as Company*".

All Sections: Share a common "*Login*" option.

Login Page: A centralized access point for all users (students, companies, and universities) to log in and utilize the **S&C** platform services.

Main Home Page: After logging in, users are presented with various actionable options and interactions available via the Menu. A "*Feedback and Improvements*" section is available to all user types for submitting suggestions to enhance the platform.

User-Specific Features

For Students

Profile Management: Students can view and edit their personal profiles, update work experience, and attach CV via the *"Add Curriculum Vitae"* option.

Internships and Collaborations: Students can browse internships aligned with their profiles, academic backgrounds, and preferences. The *"Apply"* option allows students to submit applications for desired positions.

Application Management: Status updates for internship applications are visible in the *"Offer Management"* section. Inside, it is possible to find:

- *"Pending Application"*: Applications still open.
- *"In Review"*: Applications under evaluation.
- *"Accepted Application"*: Approved by the company's HR.
- *"Rejected Application"*: Declined by the company.

After the selection process, students see *"Accepted Offer"* or *"Rejected Offer"*, with options to *"Accept"* or *"Decline"* the internship.

Selection Phase: The interview and screening process is managed in the *"Screening and Interviews"* section.

Complaint Management: Complaints are submitted through the *"Complaint Management"* section.

Collaboration Activities: Ongoing collaborations are monitored in the *"Activity Monitoring"* section.

For Companies

Company Profile: Companies can view their profiles and track the progress of published internship opportunities.

Internship Offer Optimization: Suggestions for improving internship postings are available in the *"Offer Management"* section.

Publish Internship Offers: New internship offers can be added via *"Publish Internship Offers"*, where deadlines can be set, modified, or extended. Companies can view the total number of applications received and directly access applicant profiles.

Application Management: Received applications are managed in the *"Received Applications"* section, where companies can accept or reject applications using *"Accept Application"* or *"Reject Application"* options. The interview and selection process is carried out in the *"Screening and Interviews"* section.

Final Selection: HR personnel can finalize internship selections using *"Accept Selection"* or *"Reject Application"*. The outcome is communicated to the applicant with their acceptance or rejection status.

Complaint Management: Complaints are handled through the *"Complaint Management"* section.

Collaboration Monitoring: Active collaborations are tracked in the *"Activity Monitoring"* section.

For Universities

Student and Company Management: Universities can access a list of students and companies involved in internships, categorized by school, department, faculty, and program.

Notifications: Notifications are triggered when complaints are raised by students or companies, allowing the university to respond promptly. Notifications are linked to the *"Complaint Management"* section.

Internship Tracking: Universities can monitor internship progress, with updates provided by students and companies to stay informed about the internship's status and developments.

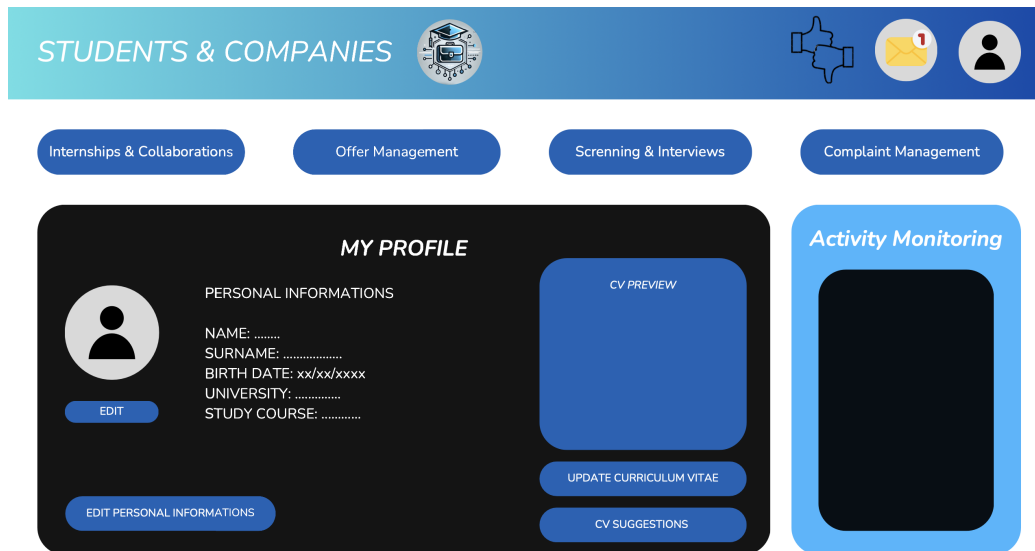


Figure 4: Student HomePage

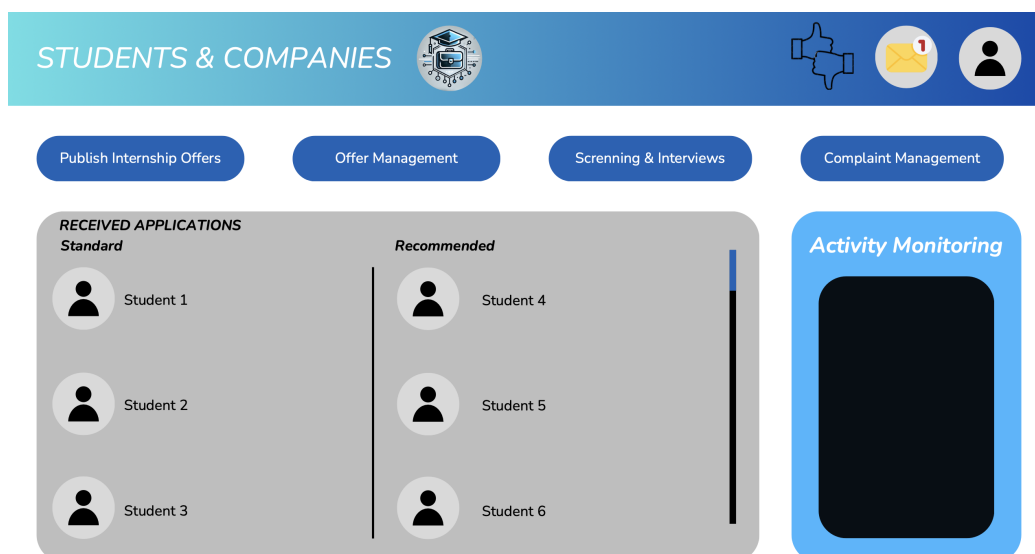


Figure 5: Company HomePage

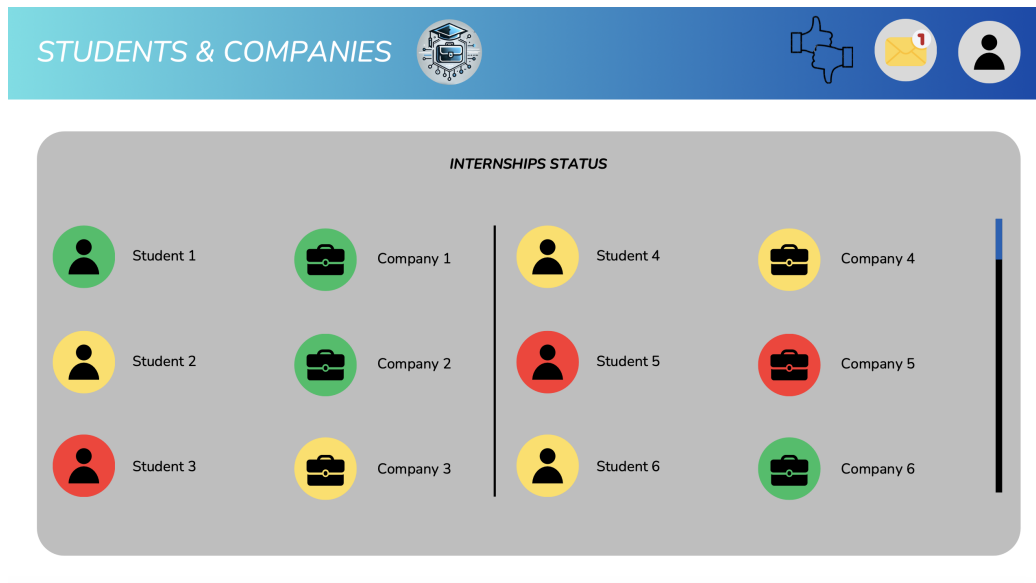


Figure 6: University HomePage

3.1.2 Hardware Interfaces

This platform does not require any special hardware to perform the functions it offers. Since it can be used as a website, users need at least a computer or mobile phone.

3.1.3 Software Interfaces

Automatic System of S&C

- The **Automatic CV Analysis Interface**, leveraging natural language processing (NLP) algorithms and machine learning techniques, reviews résumés uploaded by students. This interface provides personalized suggestions to improve CV quality, helping students present themselves optimally to companies.
- The **Matching and Recommendation Interface**, a key component that connects students with internship opportunities. It uses advanced algorithms to analyze profile and offer data, communicates with the database to retrieve necessary information, and generates relevant recommendations.
- The **Automated Questionnaire Assessment Interface**, which performs analytical operations for static evaluation of multiple-choice questions within questionnaires sent to students during the selection phase.

The **Support and Feedback Interface**, enabling users to submit assistance requests, provide feedback, or report issues. This interface integrates with ticketing or email systems to effectively manage communications and ensure timely user support.

The **Complaint Management Interface**, specifically designed to handle user reports about problems or dissatisfaction encountered during internships.

3.1.4 Communication Interfaces

Since the platform is a **Web-Based Application**, users must have an internet connection to access its features, such as searching for internships, receiving applications, getting notifications, submitting complaints, and obtaining improvement suggestions. As outlined above, the system requires automatic email dispatch in various contexts. To handle this, it leverages the **API** of the selected automated email delivery service.

3.2 Functional Requirements

This section outlines a series of functional requirements necessary to achieve the “Goals” mentioned in Section 1.1, through the “Assumptions, Dependencies, and Constraints” described in Section 2.4.

1.Registration and Identification

<i>ID</i>	<i>Description</i>
R1	The system must allow students to register using truthful, detailed personal information.
R2	The system must allow companies to register on the platform using contact information and details about the company’s mission.
R3	The system must allow universities to register on the platform after entering into an agreement with S&C.
R4	The system must enable students to verify their identity to access the system through the sign-in operation.
R5	The system must enable companies to verify their identity to access the system through the sign-in operation.
R6	The system must enable universities to verify their identity to access the system through the sign-in operation.
R7	The system must restrict access to the platform’s functionalities for unauthenticated users.

2.User Profile Management

<i>ID</i>	<i>Description</i>
R8	The system must allow students to update their personal profile.
R9	The system must allow students to upload their CV in PDF format.
R10	The system must allow students to view and update their CV, including displaying the date it was last updated.

<i>ID</i>	<i>Description</i>
R11	The system must provide suggestions for improving the students' CVs.
R12	The system must allow companies to create, modify, and publish internship offers.
R13	The system must provide suggestions for improving internship offers to companies.
R14	The system must allow companies to modify the contact information of the internship coordinator.
R15	The system must allow universities to monitor the progress of internships.

3. Matching and Recommendation System

<i>ID</i>	<i>Description</i>
R16	The system must perform matching between student profiles and internship offers.
R17	The system must allow students to receive recommendations for relevant offers.
R18	The system must allow companies to receive suggestions for suitable students for their offers.
R19	The system must allow users and companies to provide feedback on the recommendations.

4. Application and Selection Management

<i>ID</i>	<i>Description</i>
R20	The system must allow students to apply for internship offers through the platform.
R21	The system must allow students to withdraw their application.
R22	The system must allow companies to view the applications received.

<i>ID</i>	<i>Description</i>
R23	The system must allow companies to evaluate candidates through structured questionnaires.
R24	The system must collect responses to the questionnaires and present the results to companies.
R25	The system must allow companies to communicate the outcome of the applications.
R26	The system must allow students to accept or reject the offers received.

5.Internship Monitoring and Management

<i>ID</i>	<i>Description</i>
R27	The system must update the internship status to “Active” after acceptance.
R28	The system must allow universities, students, and companies to monitor the progress of the internship.
R29	The system must allow students and companies to report issues or complaints.
R30	The system must notify the university corresponding to the student when a complaint is present.
R31	The system must allow the university to contact the student and the company to resolve any issues arising during the internship.
R32	The system must allow companies to provide feedback on the students’ performance during and after the internship.
R33	The system must allow students to provide feedback on their internship experience.
R34	The system must manage the early termination of the internship, if requested by the university, through a defined procedure involving all parties.

6.Communication and Notifications

<i>ID</i>	<i>Description</i>
R35	The system must provide an internal messaging feature for notifications and important alerts.
R36	The system must be able to send email notifications for important events.
R37	The system must integrate with external services for automated email sending.

3.2.1 Mapping on goals

<i>Goal</i>	<i>Selected Requirements</i>	<i>Domain Assumptions</i>
G1: Facilitate connections between students, universities, and companies for internships.	R1, R2, R3, R4, R5, R6, R7, R8, R9, R10, R12, R14, R16, R17, R18, R20, R22, R35, R36, R37	A01, A02, A03, A04, A05, A07
G2: Improve the quality of applications and offers.	R8, R9, R10, R11, R12, R13, R35	A02, A04, A06, A07
G3: Effectively manage the selection process.	R20, R21, R22, R23, R24, R25, R26, R35, R36, R37	A02, A04, A05, A07
G4: Monitor the execution of internships.	R3, R15, R27, R28, R30, R31, R32, R33, R34, R35	A01, A05, A07
G5: Promote collaboration between students, companies, and universities.	R14, R15, R28, R30, R31, R32, R33, R35, R36, R37	A01, A02, A03, A04, A05, A07
G6: Improve the quality of matches.	R16, R17, R18, R19	A01, A04, A05, A07
G7: Support stakeholders in problem resolution.	R29, R30, R31, R34, R35	A01, A07

GOAL1: The selected requirements satisfy the first goal by enabling the creation of an integrated platform connecting Students, Companies, and Universities, allowing them to easily search for, post, and manage internships. Registration and authentication ensure secure access to the system. The management of offers and profiles provides the necessary information for effective matching between students and companies. Additionally, the matching and recommendation functionality facilitates the pairing process between Companies and Students. A key feature lies in the application process and internal/external messaging, enabling direct interactions between parties while keeping them informed and engaged.

GOAL2: The selected requirements provide tools that help students in improving their CVs and companies enhance their internship offers, making them better suited and aligned with their respective needs. Personalized suggestions and the ability to update profiles and offers ensure that informations remain high-quality and up-to-date, increasing the likelihood of successful automated matching.

GOAL3: The selected requirements shape the student selection process. They structure the stages of application, evaluation, and communication of results. Moreover, they enable students to manage their applications and provide companies with suitable tools to evaluate them effectively.

GOAL4: The selected requirements enable and ensure that Universities can supervise internships, guaranteeing that companies comply with previously established agreements and that students receive adequate training. A key aspect is communication through feedback, which allows continuous monitoring and management of the ongoing situation, enabling timely resolution of potential issues.

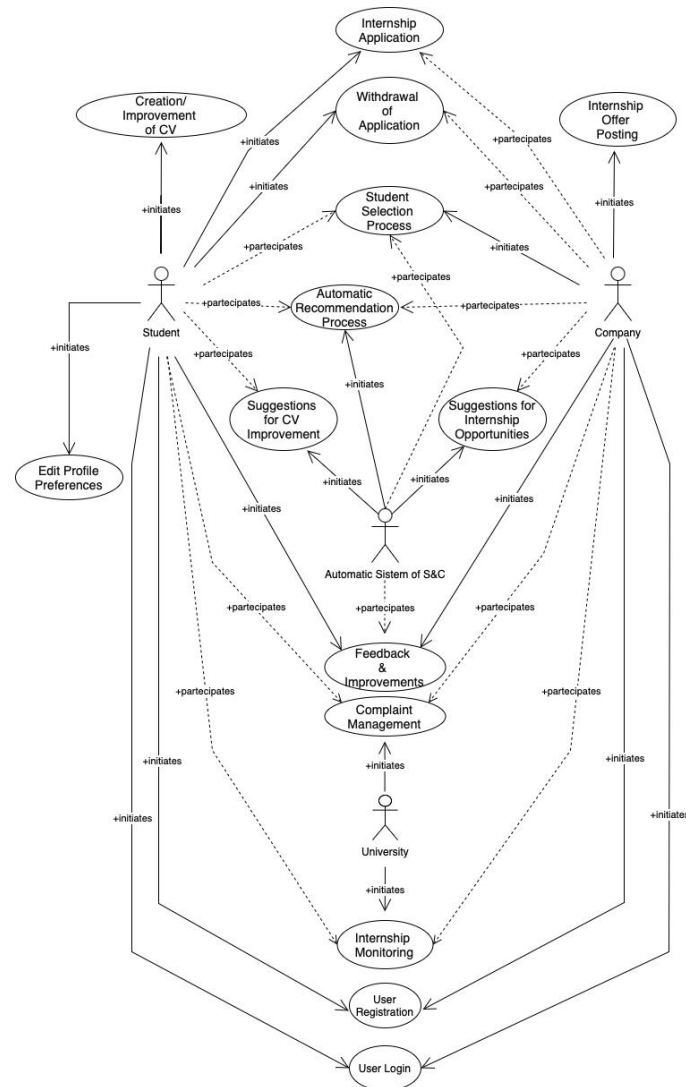
GOAL5: The selected requirements create an environment that fosters communication and cooperation between students, companies, and universities, developing an integrated system to facilitate the internship management process.

GOAL6: The selected requirements provide and implement advanced matching and recommendation mechanisms, utilizing feedback from students and companies to increase the likelihood of successful student-company pairings.

GOAL7: The selected requirements offer mechanisms for reporting and resolving issues, ensuring that students, companies, and universities can communicate easily and manage problems promptly and appropriately.

3.2.2 Use case diagrams

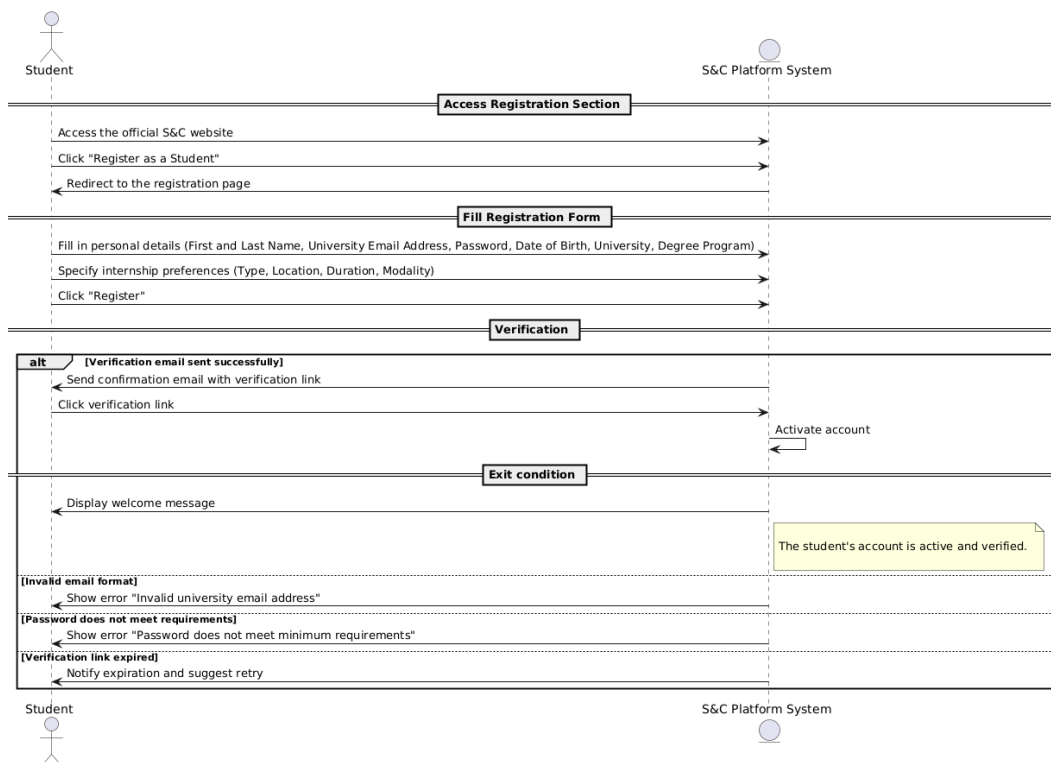
In this subsection, we will observe through **UML Use Case diagrams** every type of interaction that **Student**, **Company**, and **University** have with the **S&C** platform. The analysis will focus primarily on fundamental actions, which are considered delicate due to their importance, such as actor registration in the system, CV upload by students, modifications within actor profiles, and the uploading of internship offers by companies. Additionally, we will address the handling and management of complaints, monitoring the progress of internships, and feedback and advice both within and outside the platform.



THE STUDENT REGISTERS ON THE PLATFORM	
ID	UC1
Actor	Student
Entry Conditions	The student does not have an S&C account.
Flow of Events	1.The student accesses the official Students&Companies (S&C) platform website. 2.The student selects the registration page by clicking on the "<i>Register as Student</i>" button. 3.The system redirects the student to the registration page. 4.The student fills in the required fields: - Full Name - University Email Address - Password (to create and confirm) - Year of Birth - University Name - Degree Program 5.The student specifies initial preferences to personalize their experience: - Type of internship (e.g., curricular, extracurricular, paid/unpaid) - Desired geographic location - Maximum or minimum duration of the internship - Mode (on-site, remote) 6.The student clicks on the "<i>Register</i>" button. 7.The student receives a confirmation email and clicks on the verification link. 8.The system activates the student's account and shows a welcome message.
Exit Conditions	The student's account is active and verified.

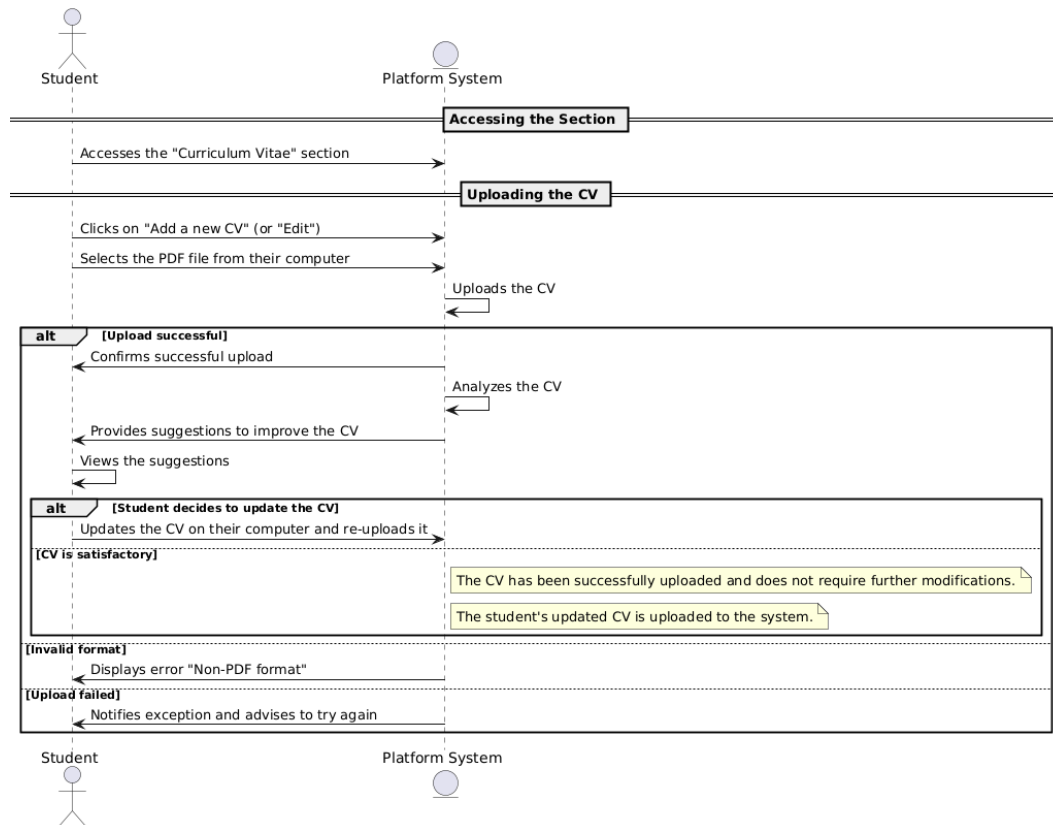
Exceptions	- If the email is not a valid institutional email from a registered university on the platform, the system does not allow registration. -If the password does not meet the minimum requirements (e.g., length, complexity), the system shows an error message. -If the email verification link is not clicked within the time limit, the system temporarily disables the registration and requests a new attempt.
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SEQUENCE DIAGRAM



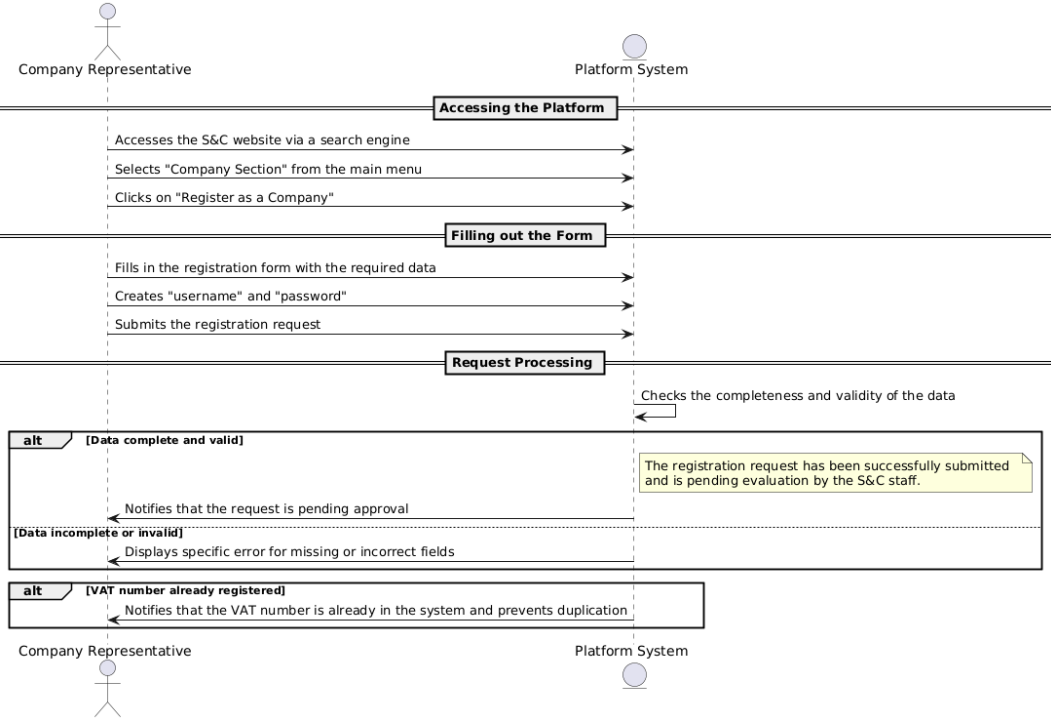
STUDENT'S CV CREATION AND IMPROVEMENT	
ID	UC2
Actor	Student
Entry Conditions	- The student is registered and authenticated. - The student has not yet uploaded their CV or wishes to update it.
Flow of Events	1.The student accesses the “<i>Curriculum Vitae</i>” section in their profile. 2.Clicks on “<i>Add a new CV</i>” or <i>Edit</i>” if updating a previous CV. 3.Selects the PDF file of their CV. 4.The system uploads the CV and confirms the success. 5.The system analyzes the CV and provides improvement suggestions. 6.The student reviews the suggestions and decides whether to modify it. 7.If needed, the student updates the CV on their PC and reuploads it.
Exit Conditions	The updated CV is uploaded to the system.
Exceptions	-If the uploaded CV is not in PDF format, the system raises an error. -If a loading error occurs, the system notifies the student and suggests him to try again.

SEQUENCE DIAGRAM



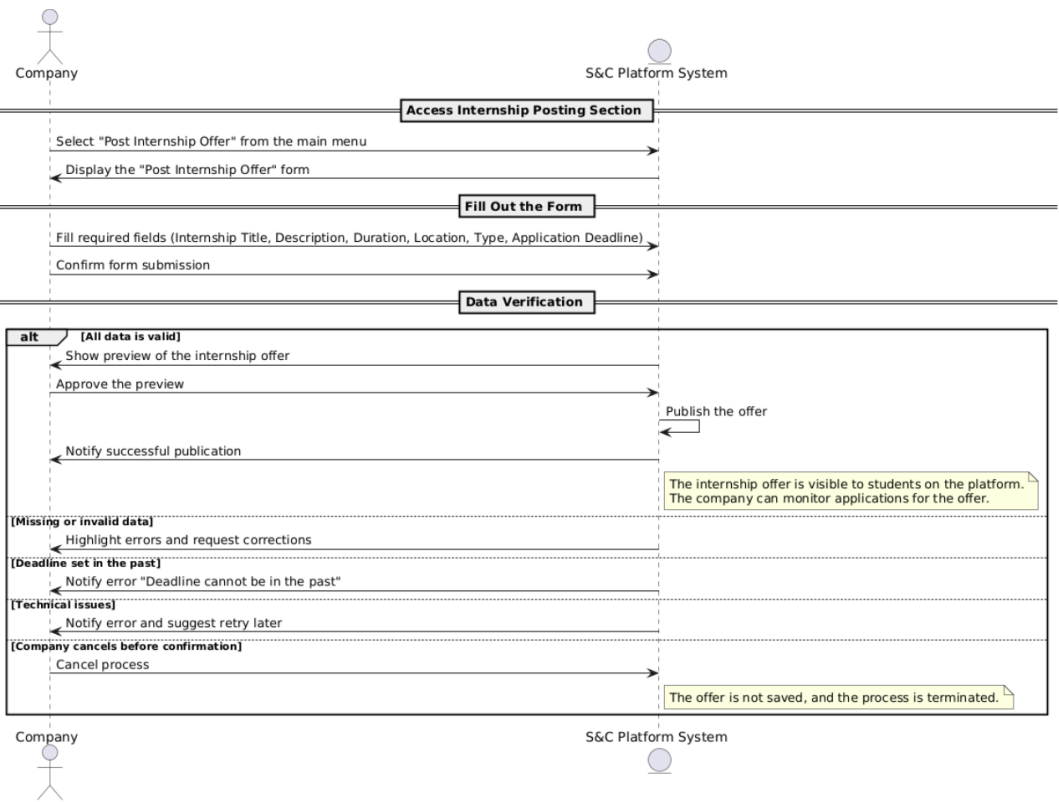
COMPANY REGISTRATION	
ID	UC3
Actor	Company Representative
Entry Conditions	- The company is not registered on the S&C platform. - The representative has access to an internet-connected device.
Flow of Events	1.The company representative accesses the S&C website via a search engine. 2.They select the "<i>Company Section</i>" page from the menu. 3.They click on "<i>Register as a Company</i>". 4.They fill out the registration form with the required information. 5.They create a username and password for future access. 6.They submit the registration request. 7.The system notifies them that the request is pending approval.
Exit Conditions	The registration request is submitted and is pending review by the S&C staff.
Exceptions	-If the entered data is incomplete or invalid, the system displays specific errors. -If the IVA number is already registered, the system informs the representative and prevents duplication.

SEQUENCE DIAGRAM



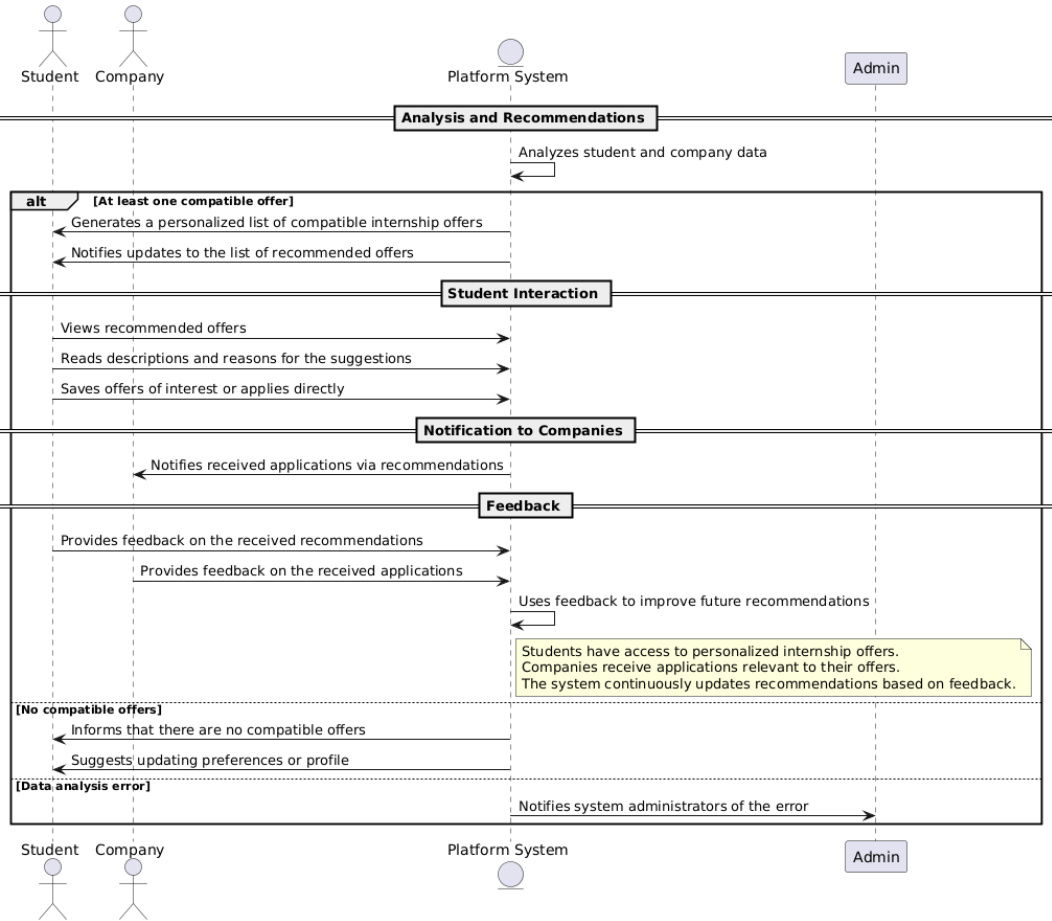
POSTING INTERNSHIP POSITIONS	
ID	UC4
Actor	Company
Entry Conditions	- The company has an active and verified account on the Students&Companies (S&C) platform. - The company is logged into the platform.
Flow of Events	1.The company selects the <i>"Publish Internship Offer"</i> option from the main menu. 2.The company completes the following required fields on the "Publish Internship Offer" page: - Internship title. - Internship description. - Expected duration. - Internship location. - Type of internship (curricular/extracurricular). - Application deadline. 3.The company confirms the form submission. 4.The system verifies the entered data and displays a preview of the offer. 5.The company approves the preview. 6.The system publishes the offer, making it visible to registered students.
Exit Conditions	- The internship offer is visible to students on the platform. - The company can monitor applications associated with the offer.
Exceptions	- Mandatory fields in the form are incomplete or invalid: the system highlights the errors and requests corrections. - The application deadline is set to a past date: the system notifies the error and requests a correction. - Technical issues prevent the publication: the company receives an error message and is asked to try again later. - The company cancels the publication before approving the preview: the offer is not saved, and the process is terminated.

SEQUENCE DIAGRAM

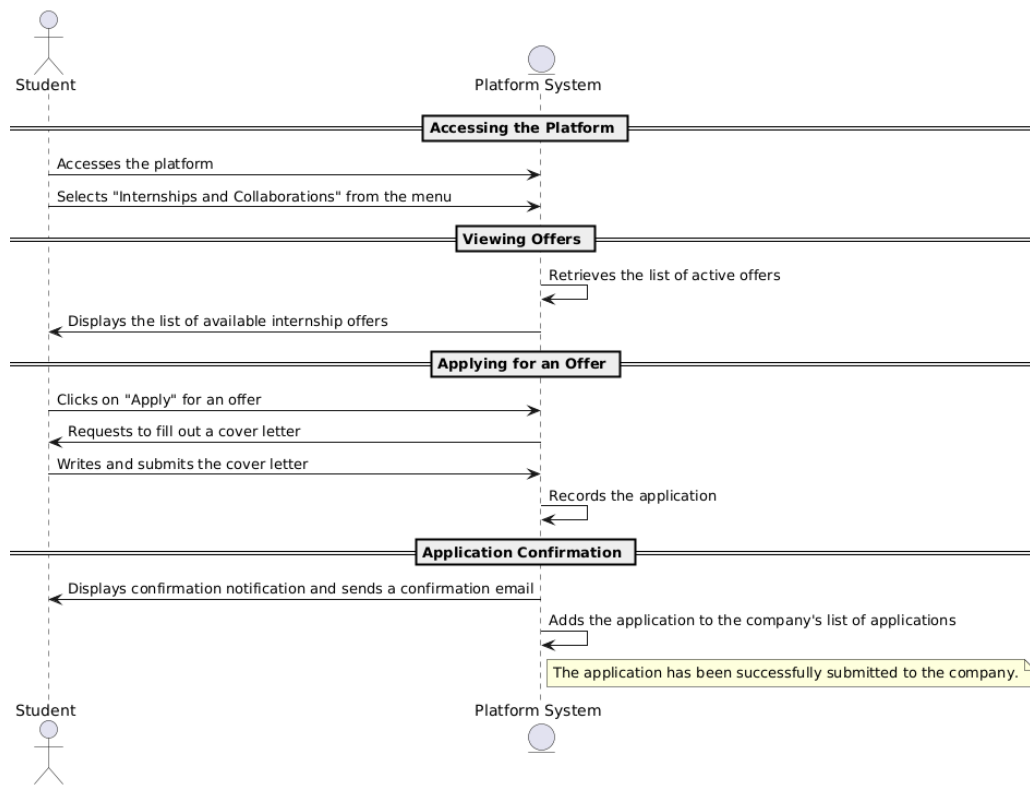


RECOMMENDATION AND MATCHING SYSTEM (MATCHING COMPANIES AND STUDENTS)	
ID	UC5
Actor	Student, Company
Entry Conditions	- Students and companies have completed their profiles on the platform and are properly registered. - Internship offers have been published by companies.
Flow of Events	1.The system analyzes the data provided by students and companies using AI and machine learning algorithms. 2.For each student, the system generates a personalized list of compatible internship offers. 3.The system notifies students of updates to their recommended internship list. 4.Students can view the recommended offers, read detailed descriptions, and understand the reasoning behind each suggestion. 5.Students can save offers of interest or apply directly through the platform. 6.The system notifies companies of received applications, including those submitted via recommendations. 7.Users can provide feedback on the recommendations received. 8.Companies can provide feedback on the applications received via recommendations. 9.The system leverages feedback to improve future recommendations.
Exit Conditions	-Companies receive applications relevant to their offers. -Students have access to personalized internship offers. -The system continuously updates recommendations based on feedback.
Exceptions	If no compatible offers are found, the system informs the student and suggests updating preferences or profile informations.

SEQUENCE DIAGRAM



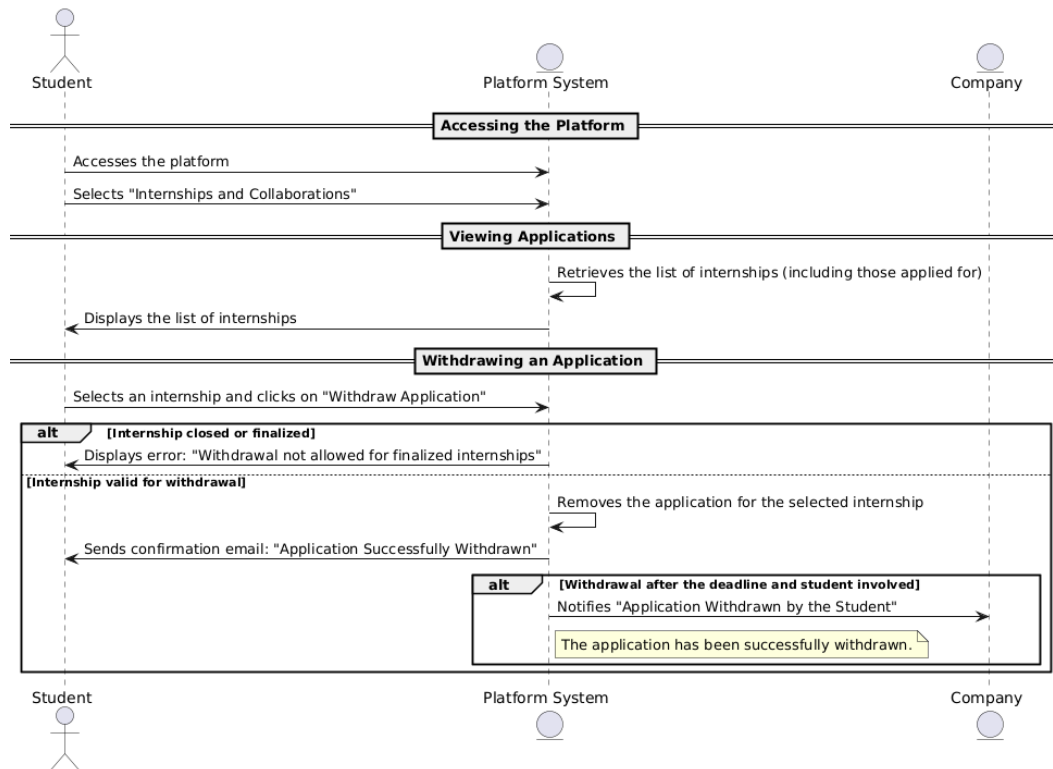
STUDENT'S SPONTANEOUS APPLICATION	
ID	UC6
Actor	Student
Entry Conditions	-The student is registered, authenticated, and has uploaded a CV on the platform. -Active internship offers are available on the S&C portal.
Flow of Events	1.The student logs into the platform and selects "<i>Internships and Collaborations</i>" from the menu. 2.They view the list of currently available internship offers. 3.They identify an offer of interest and click on "<i>Apply</i>". 4.The system prompts them to write a brief cover letter. 5.After completing the letter, the student clicks "<i>Submit Application</i>". 6.The system confirms the submission by displaying a notification and sending a confirmation email. 7.The application is added to the list of applications received by the company.
Exit Conditions	-The student's application is successfully submitted to the company. -The student receives confirmation of the submission.



SEQUENCE DIAGRAM

INTERNSHIP WITHDRAWAL AND APPLICATION CANCELLATION PROCESS	
ID	UC7
Actor	Student
Entry Conditions	- The student is authenticated on the S&C platform. -The student has submitted one or more internship applications.
Flow of Events	1.The student accesses the <i>"Internships and Collaborations"</i> section on the platform. 2.The system displays the list of available and open internships, including those the student has applied for. 3.The student selects an internship for which they want to withdraw their application. 4.The student clicks the <i>"Withdraw Application"</i> button. 5.The system processes the request and removes the application for the selected internship. 6.The system sends a confirmation email to the student with the message: <i>"Application Successfully Withdrawn"</i>. 7.If the withdrawal occurs after the application deadline and the student is involved in subsequent phases, the system notifies Company with the message: <i>"Application Withdrawn by Student"</i>.
Exit Conditions	-The application is successfully withdrawn for the selected internship. -The student receives a confirmation email. -Company is notified, if applicable.
Exceptions	If the student attempts to withdraw an application for a closed or finalized internship, the system displays an error: <i>"Withdrawal not allowed for finalized internships"</i> .

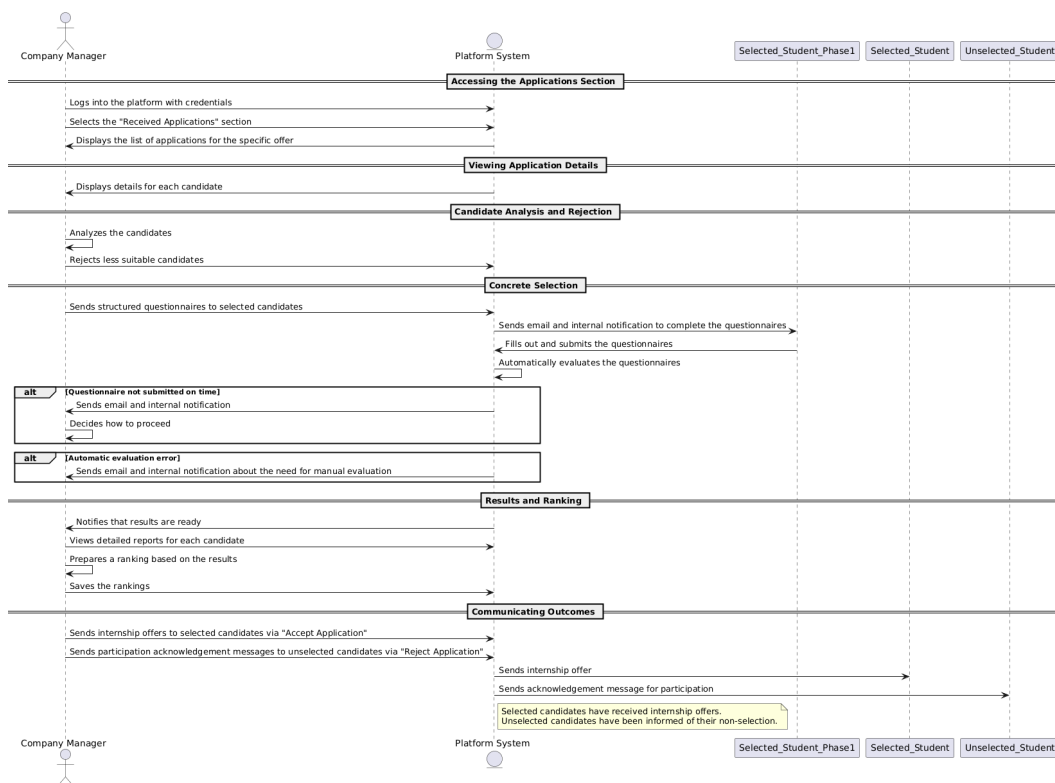
SEQUENCE DIAGRAM



INTERNSHIP APPLICATION REVIEW AND SELECTION PROCESS	
ID	UC8
Actor	Company
Entry Conditions	-The application period for the internship has ended. -The company has received applications for a specific internship offer.
Flow of Events	1.The company manager logs into the platform using his credentials. 2.He navigate to the “<i>Received Applications</i>” section. 3.The system displays the list of applications for the specific internship offer. 4.For each applicant, the system provides the following details: - Full Name - University of Enrollment - Field of Study - Student’s CV - Compatibility Score 5.The company manager reviews the applicants and excludes less suitable candidates based on predefined criteria, sending Participation Acknowledgments. 6.The concrete selection phase begins. 7.For shortlisted candidates, the manager sends structured questionnaires via the platform. 8.Students receive email and system notifications, prompting them to complete the questionnaires to proceed in the selection process. 9.Once students submit their responses, the system automatically evaluates the questionnaires. 10.The manager receives a notification when the evaluation results are ready. 11.He review detailed reports for each candidate. 12.A ranking is compiled based on the results. 13.Internship offers are sent to selected candidates via the “<i>Accept Application</i>” button. 14.Participation Acknowledgments are sent to non-selected candidates via the “<i>Reject Application</i>” button.
Exit Conditions	- Selected candidates have received internship offers. - Non-selected candidates have been notified about their non-selection.

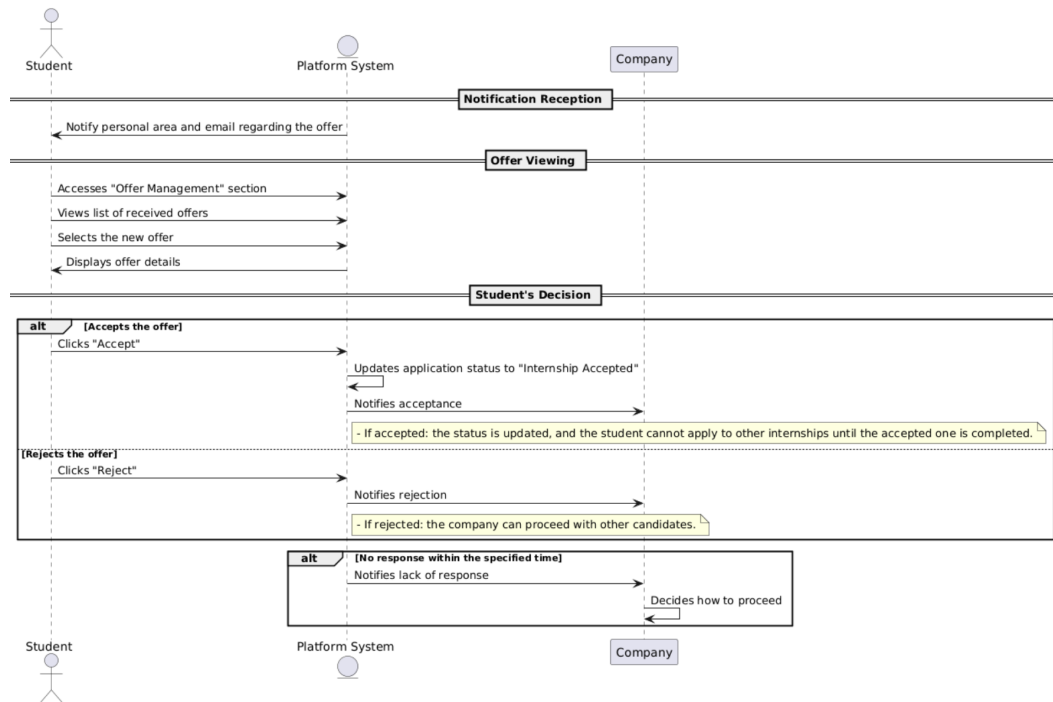
Exceptions	- If a student fails to complete the questionnaire within the allotted time, the system notifies the company via email and internally, allowing the company to decide how to proceed. - If the system detects potential errors in the automatic evaluation, it notifies the company via email and internally, requesting a manual review.
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SEQUENCE DIAGRAM



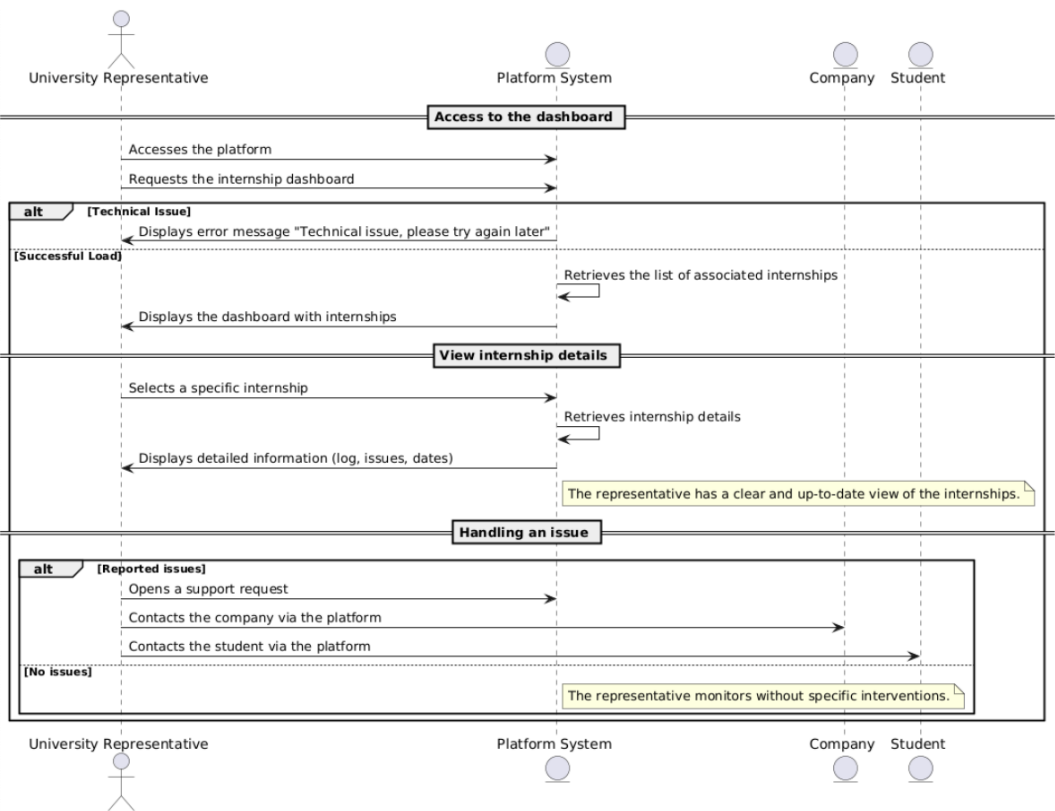
THE STUDENT HAS BEEN SELECTED FOR AN INTERNSHIP IN A COMPANY	
ID	UC9
Actor	Student
Entry Conditions	-The student is already authenticated on the platform. -The student has received an internship offer.
Flow of Events	1.The student receives a notification in their personal area and via email about the offer. 2.They navigate to the “<i>Offer Management</i>” section. 3.They view the list of received offers and select the new offer. 4.They review all the offer details, including: - Start and end dates - Working hours - Responsibilities and expected tasks - Compensation and benefits (if applicable) 5.The student decides whether to accept or decline the offer. 6.They click “<i>Accept</i>” to accept the offer. 7.The system updates the application status to “<i>Internship Accepted</i>” and notifies the company. 8.If the student declines, they click “<i>Decline</i>”, and the system notifies the company of the rejection.
Exit Conditions	- If accepted, the status is updated, and the student cannot apply for other internships until the accepted one is completed. -If declined, the company can proceed with other candidates.
Exceptions	If the student does not respond within the specified period, the system notifies the company, which decides how to proceed.

SEQUENCE DIAGRAM



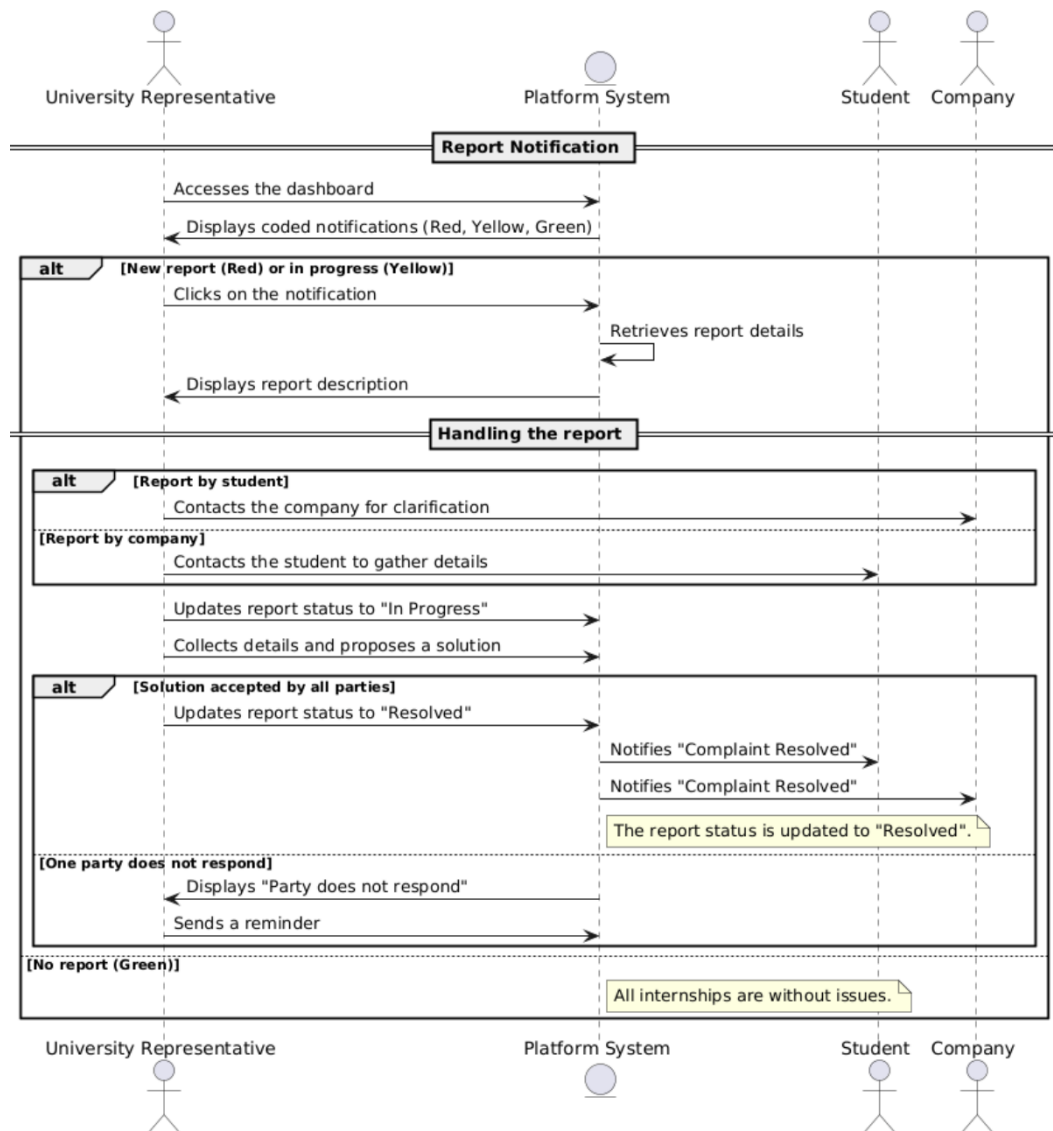
MONITORING OF INTERNSHIPS BY THE UNIVERSITY	
ID	UC10
Actor	University
Entry Conditions	- The university is registered on the platform with an institutional account and has completed the registration process. - The student undertaking the internship is associated with the university. - The company has shared the internship details with the university through the platform.
Flow of Events	1.The system displays a dashboard containing a detailed list of internships associated with the university's students. For each internship, the main information includes: - Internship title. - Student's name. - Company name. - Status of the internship. 2.The university representative can click on a specific internship to view detailed information, including: - The student's activity log. - Any issues reported by the student or the company. - Start and end dates of the internship. 3.If the representative identifies a problem or issue, they can choose to: - Open a support request. - Contact the student or company directly through the platform.
Exit Conditions	- The university has a clear and updated overview of the internships associated with its students. - The university representative can take action to address reported issues or monitor internship activities.
Exceptions	A technical issue occurs while loading the dashboard: an error message is displayed, and the representative is prompted to try again later.

SEQUENCE DIAGRAM



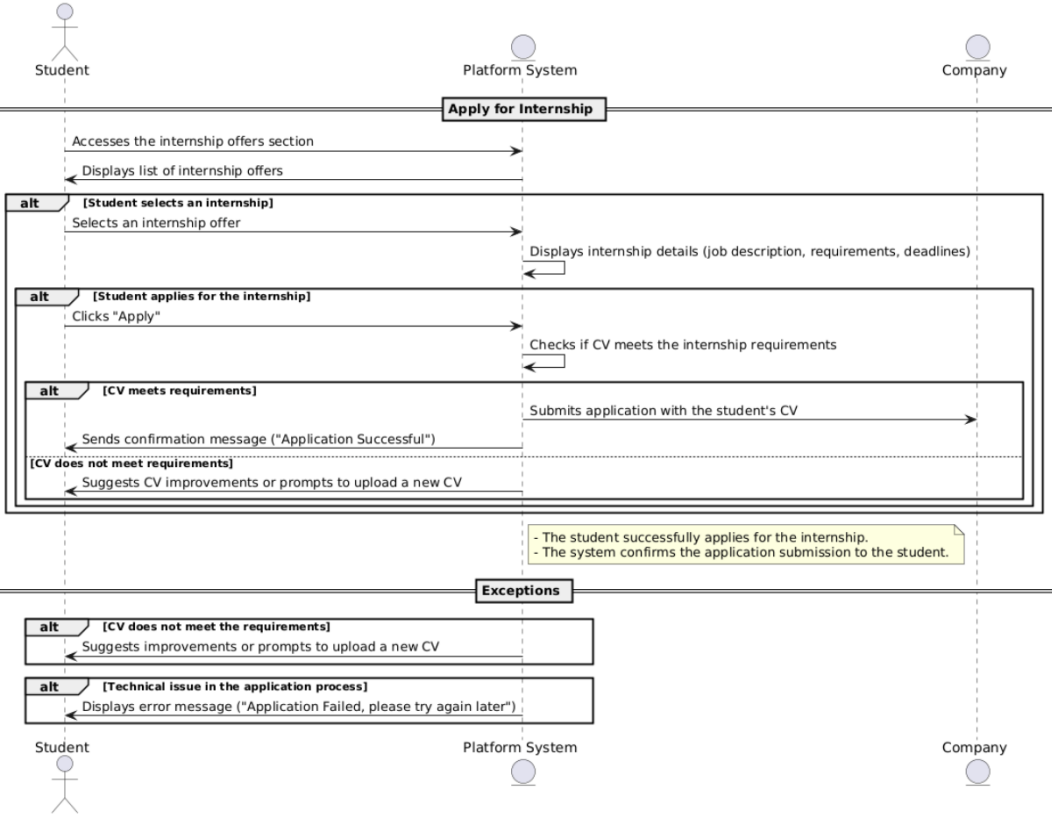
MANAGEMENT OF REPORTS AND COMPLAINTS BY THE UNIVERSITY	
ID	UC11
Actor	University
Entry Conditions	- The university is registered and has access to the platform. - There is a report (from a student or a company) related to an ongoing internship.
Flow of Events	1.A color-coded notification is visible on the dashboard: - Red: New report not yet viewed. - Yellow: Report in progress, not yet resolved. - Green: Internships without issues. 2.By clicking on the notification, the university representative accesses the "<i>Complaint Management</i>" section, where they can read the description of the submitted report: - If the student submitted the report: the university can contact the company for clarification. - If the company submitted the report: the university can contact the student to gather details. 3.Using the platform, the university can initiate communication with the other involved party, gather additional information, and propose a solution. 4.Once the complaint is resolved, the university representative updates the status of the report to "<i>Resolved</i>", automatically notifying all involved parties (student or company) via email.
Exit Conditions	The report is resolved, and the status is updated to " <i>Resolved</i> ".
Exceptions	- If one of the parties does not respond to clarification requests, the university sends a reminder via the system.

SEQUENCE DIAGRAM



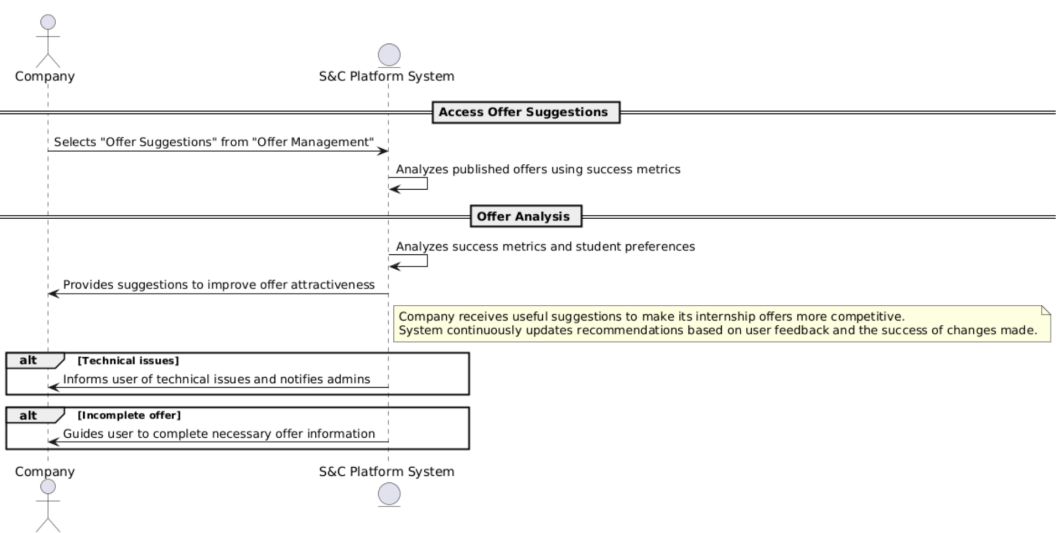
CV IMPROVEMENTS FOR STUDENTS	
ID	UC12.1
Actor	Student
Entry Conditions	- The student is registered on the platform and has uploaded a CV to their profile. - Sufficient data is available for analysis, such as CVs from other users.
Flow of Events	1.The student selects the "CV Suggestions" section from the main menu. 2.The system analyzes the student's CV using analysis algorithms based on: - Comparison with the skills required in internship offers. - Common experiences of students who successfully obtained internships. 3.The system generates personalized suggestions to improve the CV, such as: - Adding relevant technologies or skills for specific sectors. - Optimizing the layout or including an initial skills summary. 4.The student views the suggestions.
Exit Conditions	- The student receives detailed feedback to improve their CV. - The system continuously updates recommendations based on user feedback and the success of changes made.
Exceptions	- Technical issues in data analysis: the system informs the user and reports the error to administrators. - The student has not uploaded a CV: the system prompts the user to upload a CV.

SEQUENCE DIAGRAM



SUGGESTIONS FOR COMPANY	
ID	UC12.2
Actor	Company
Entry Conditions	- The company has published at least one internship offer on the platform. - Sufficient data is available for analysis.
Flow of Events	1.The company selects the "<i>Offer Suggestions</i>" section in the "<i>Offer Management</i>" section. 2.The system analyzes the published offers using: - Success metrics from previous offers. - Analysis of student preferences and behaviors on the platform. 3.The system provides suggestions to improve the attractiveness of the offers, such as: - Specifying benefits or growth paths. - Clarifying technical requirements and expectations.
Exit Conditions	- The company receives useful suggestions to make its internship offers more competitive. - The system continuously updates recommendations based on user feedback and the success of changes made.
Exceptions	- The system fails to identify suggestions for the offer: the user is notified that available data is insufficient. - Technical issues in data analysis: the system informs the user and reports the error to administrators.

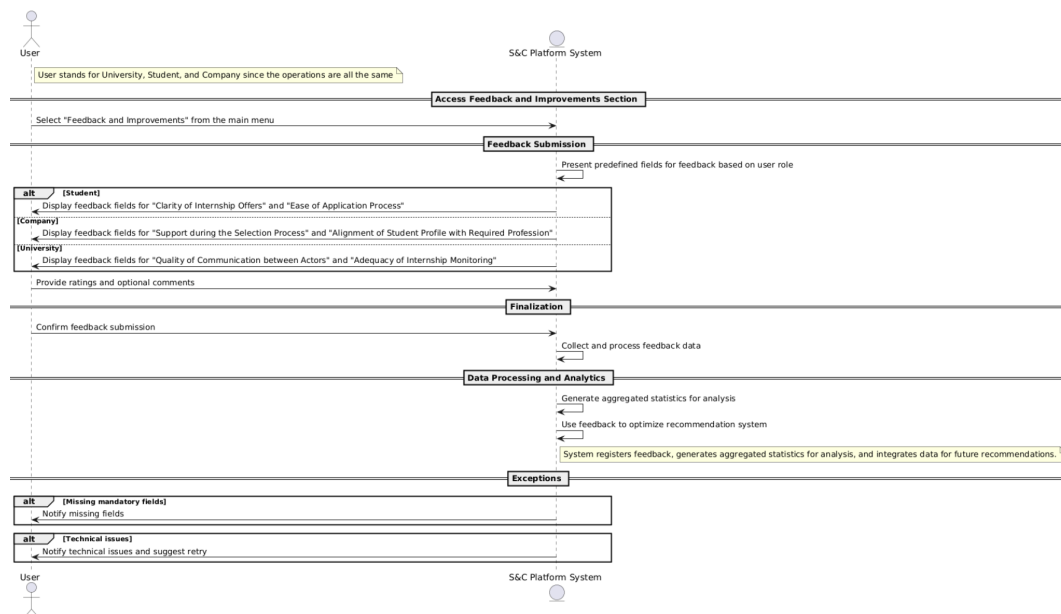
SEQUENCE DIAGRAM



FEEDBACK AND STATISTICS COLLECTION	
ID	UC13
Actor	Student Company University
Flow of Events	- Users (Students, Companies, Universities) have access to the S&C platform. - The "Feedback and Improvements" section is active.
Entry Conditions	1.From the main menu, users select the "Feedback and Improvements" option. 2.The platform presents a section with predefined fields to collect evaluations on various aspects, differentiated based on the user role: - Student: — <i>"Clarity of Internship Offers"</i>. — <i>"Ease of Application Process"</i>. -Company: — <i>"Support During the Selection Process"</i>. — <i>"Alignment of Student Profile with Required Profession"</i>. -University: — <i>"Quality of Communication Between Actors"</i>. — <i>"Adequacy of Internship Monitoring"</i>. 3.Users provide ratings through rating systems and can add optional comments for additional details. 4.Once the feedback is complete, the user confirms submission. 5.The system collects the data and processes it to generate: -Aggregated statistics useful for analysis and improvements. -Information to optimize the recommendation system and improve the user experience. 6.The results are integrated into the system's learning database to refine future recommendation and matching processes.
Exit Conditions	-The system records the feedback provided by users. -Aggregated statistics are generated for subsequent analysis.

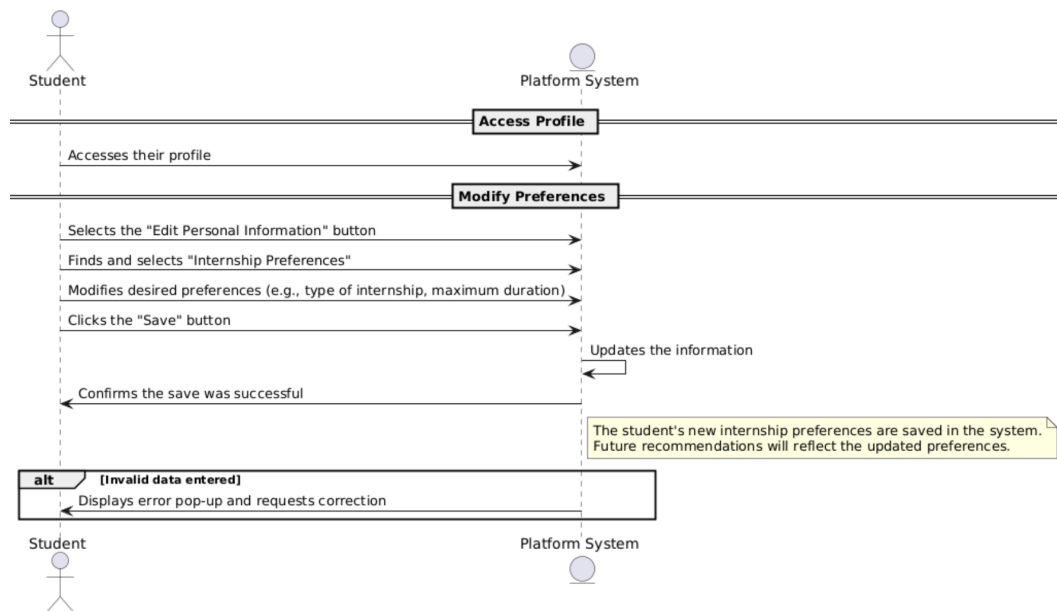
Exceptions	- If a user does not complete all required fields, the system highlights the missing sections before accepting the feedback. - In case of technical issues, the system notifies the user and suggests retrying later.
------------	--

SEQUENCE DIAGRAM



MODIFY STUDENT PROFILE TO CHANGE INTERNSHIP PREFERENCES	
ID	UC14
Actor	Student
Entry Conditions	- The student is registered and authenticated on the platform. - The student wishes to update previously set internship preferences.
Flow of Events	1.The student accesses their profile in the application. 2.They navigate to the “<i>Edit Personal Information</i>” section. 3.They locate and select “Internship Preferences”. 4.The student modifies the desired preferences, such as type of internship and maximum duration. 5.They click the “<i>Save</i>” button to confirm the changes. 6.The system updates the information and confirms the successful save.
Exit Conditions	-The student’s new internship preferences are saved in the system. -Future recommendations will reflect the updated preferences.
Exceptions	If the entered data is invalid (e.g., negative duration), the system displays an error pop-up and prompts for correction.

SEQUENCE DIAGRAM



3.2.3 Mapping on requirements

In this section, we will provide an overview of the relationships between use cases and requirements. The purpose of this overview is to verify and demonstrate how the system will fulfill these requirements.

<i>Use Case ID</i>	<i>Requirement IDs</i>
UC1	R1, R3, R36
UC2	R4, R7, R9, R10, R11
UC3	R2, R36
UC4	R5, R7, R12
UC5	R16, R17, R18, R19, R20, R35
UC6	R4, R7, R9, R12, R20, R36
UC7	R4, R7, R20, R21, R36
UC8	R5, R7, R12, R16, R20, R22, R23, R24, R25, R26, R35, R36
UC9	R4, R7, R12, R26, R27, R35, R36
UC10	R29, R30, R31, R35
UC11	R29, R30, R31, R35, R36
UC12.1	R4, R7, R9, R10, R11, R19
UC12.2	R5, R7, R12, R13, R19
UC13	R4, R5, R6, R7, R19
UC14	R4, R7, R8, R17

3.3 Performance Requirements

Details on performance expectations and targets for the product.

3.4 Design Constraints

In this section we'll describe some constraints of the design processes that will take place after the requirement analysis phase of our developing process.

3.4.1 Standards Compliance

Some sensitive data is needed to be stored in S&C's DB, such as user's personal data. Consequently, when a user logs in for the first time, he will be asked for the permission to handle these data. The platform will be compliant to the European Union's General Data Protection Regulation (GDPR) to ensure that the data of its users will be stored and used in accordance to the law.

3.4.2 Hardware Limitations

Customer device

1. Stable internet connection (3G or more for mobile devices)
2. A supported version of one of the most common web browsers (to access to the web portal)

Server

1. Stable and cabled internet connection
2. Enough memory and computational power to handle thousands parallel requests
3. Enough disk space to store the DB

3.5 Software System Attributes

3.5.1 Reliability

The service is required to be fault tolerant, in order to guarantee its continuity. It should minimize the occurrence of failures and errors, providing accurate and consistent results to users, it is also necessary to schedule a periodic backup to prevent data loss.

3.5.2 Availability

The system will be designed to maintain high availability, aiming for minimal downtime. This will be achieved through redundant server configurations, load balancing and other design decisions that will take place during the design phase.

3.5.3 Security

Ensuring data privacy and system integrity, as stated in section (3.4.1), will have primary importance. The system will employ encryption methods and proper authentication mechanisms to safeguard user data.

3.5.4 Maintainability

The system will place a strong emphasis on ensuring ease of maintenance as one of its primary objectives. It will consist of well-commented code and proper documentation to facilitate future updates, bug fixes, and improvements.

3.5.5 Portability

The system will be designed to be accessible from all kind of devices that support any modern browser.

4 Formal Analysis Using Alloy

This section includes the formal specification and analysis of the system using the Alloy tool. This section includes a brief presentation of the main objectives driving the formal modeling activity, as well as a description of the model itself, what can be proved with it, and why what is proved is important given the problem at hand. To show the soundness and correctness of the model, this section can show some worlds obtained by running it, and/or the results of the checks performed on meaningful assertions.

```
1
2
3
4 // + - - - - - - - - - - - - - - - +
5 // |                               SIGNATURE                               |
6 // + - - - - - - - - - - - - - - - +
7
8 // Booleans
9 abstract sig Boolean {}
10 lone sig True extends Boolean {}
11 lone sig False extends Boolean {}
12
13
14 one sig SandC {
15   var universities: some University,
16   var companies: some Company,
17   var students: some Student
18 }
19 enum Status {
20   OPEN,
21   ONGOING,
22   CLOSED
23 }
24 enum ApplicationResults {
25   PENDING,
26   APPROVED,
27   REJECTED
28 }
29 // Date time , represented as an integer
30 sig DateTime {
31   time : one Int
32 }{
33   time >= 0
```

```

34 }
35 sig Email {}{
36     one this .~ username
37 }
38 sig Password {}{
39     some this .~ password
40 }
41 var sig CV{}{
42     one s: Student | this=s.cv
43 }
44 sig Internship {
45     state: one Status,
46     startDate: one Int,
47     endDate: one Int,
48     applicationDeadline: one Int,
49     company: one Company // Direct relationship with the
                           company
50 } {
51     startDate < endDate
52     applicationDeadline < startDate
53 }
54 sig InternshipApplication {
55     application: one Internship,
56     result: one ApplicationResults,
57     submissionDate: one Int
58 } {
59     submissionDate >= 0
60 }
61 sig SupportRequest {
62     student: one Student,
63     company: one Company,
64     university: one University,
65     internship: one Internship,
66     state: one Status
67 }
68 sig Feedback {
69     text: String
70 }
71 //Users
72 abstract sig User {
73     username : disj one Email,
74     password : one Password,
75     feedback: set Feedback

```

```

76 }
77 sig Company extends User {
78   internshipOffer: some Internship
79 }
80 sig University extends User{}
81 sig Student extends User {
82   var cv: disj lone CV,
83   var interApp: set InternshipApplication,
84   ongoingIntern: lone Internship, // "lone" to allow no
      ongoing internships
85   university: one University
86 }
87
88 // + - - - - - - - - - - - - - - - - - - - - +
89 // |                                     FACT                                     |
90 // + - - - - - - - - - - - - - - - - - - - - +
91 fact InternshipOfferedByOneCompany {
92   // Each Internship is offered by only one company
93   always(all disj c1, c2: Company | c1.internshipOffer
      != none and c2.internshipOffer != none implies
94     (c1.internshipOffer & c2.internshipOffer) = none)
95 }
96 fact InternshipWithOneStudent{
97   // Each Internship is done by one student
98   always(all disj s1, s2: Student| s1.ongoingIntern!=
      none and s2.ongoingIntern != none implies
99     (s1.ongoingIntern & s2.ongoingIntern) = none)
100 }
101 fact UniqueRelationships {
102   // Ensures unique CV per student
103   always(all s: Student | lone s.cv)
104 }
105 fact StudentMustHaveUniversity {
106   // Each student must have exactly one university
107   always(all s: Student | one s.university)
108 }
109 fact StudentUniversityAssociation {
110   // Ensures student and university association
      consistency
111   always(all s: Student | one s.university)
112 }
113 fact InternshipStudentCompanyConsistency {
114   // Links internships to students and their companies

```

```

115     always(all s: Student |
116         some s.ongoingIntern implies s.ongoingIntern.
            company in Company)
117 }
118 fact CVExistenceCondition {
119     // Ensures CVs belong to students
120     always(all c: CV | some s: Student | c in s.cv)
121 }
122 fact StudentMustHaveCVForInternship {
123     // Students need a CV for internships
124     always(all s: Student |
125         (some s.ongoingIntern or some s.interApp) implies
            some s.cv)
126 }
127 fact ApplicationExistenceCondition {
128     // Ensures applications belong to internships and
        students
129     always(all a: InternshipApplication |
130         some s: Student | (a in s.interApp and a.
            application in Internship))
131 }
132 fact ApplicationLinkedToInternship {
133     // Applications must link to one internship
134     always(all app: InternshipApplication | one app.
        application)
135 }
136 fact OngoingInternshipStatus {
137     // Validates internship state and student
        participation
138     always(all i: Internship | i.state = ONGOING iff some
        i.~ongoingIntern)
139 }
140 fact FeedbackLinkedToUser {
141     // Feedback must be linked to a user
142     always(all f: Feedback | some f.~feedback)
143 }
144 fact SupportRequestConsistency {
145     // Validates support request structure
146     always(all req: SupportRequest |
147         req.student.ongoingIntern = req.internship and
148         req.company = req.internship.company)
149 }
150 fact UniqueApplicationPerStudent {

```

```

151     // One application per internship per student
152     always( all s: Student, i: Internship |
153         #(s.interApp & i.~application) <= 1)
154 }
155 fact ApplicationSubmissionDateCondition {
156     // Applications must be submitted before deadlines
157     always(all a: InternshipApplication |
158         a.submissionDate < a.application.
159             applicationDeadline)
160 }
161 fact InternshipOngoingCondition {
162     // Ongoing internships must respect date constraints
163     always(all i: Internship |
164         (i.state = ONGOING) iff (i.startDate <= i.endDate
165             ))
166 }
167 fact RejectApplicationsOnInternshipStart {
168     // Reject pending applications when internship starts
169     always(all i: Internship |
170         i.state = ONGOING implies
171         all a: InternshipApplication |
172             (a.application = i and a.result != APPROVED)
173             =>
174             a.result = REJECTED)
175 }
176 fact SupportRequestTerminationCondition {
177     // Support requests end when internships close
178     always(all req: SupportRequest |
179         req.internship.state = CLOSED implies req.state =
180             CLOSED)
181 }
182 fact NoOverlapBetweenInternshipStates {
183     // Validates mutual exclusivity of internship states
184     always(no i: Internship |
185         (i.state = OPEN and (i.state = ONGOING or i.state
186             = CLOSED)) or
187         (i.state = ONGOING and (i.state = OPEN or i.state
188             = CLOSED)) or
189         (i.state = CLOSED and (i.state = OPEN or i.state
190             = ONGOING)))
191 }
192 fact SingleSupportRequestPerInternship {
193     // Only one support request per internship

```

```

187     always(all i: Internship | #(i.~internship) <= 1)
188 }
189 fact noStudentContractUniveristy{
190     always(not some s: Student | s in SandC.students and s.
191         university not in SandC.universities)
192 }
193 fact{
194     always(all s: Student | s not in SandC.students implies
195         not s.interApp.application.~internshipOffer in SandC
196         .companies)
197 }
198
199 // + - - - - - - - - - - - - - - - - +
200 // |                               PREDICATES                               |
201 // + - - - - - - - - - - - - - - - - +
202
203 pred EligibleForInternship(s: Student, i: Internship) {
204     // Checks if a student is eligible for an internship
205     some s.cv
206     i.state = OPEN
207     no i.~ongoingIntern
208 }
209
210 pred ValidSupportRequest(req: SupportRequest) {
211     // Validates a support request
212     req.internship.state = ONGOING
213     req.student.ongoingIntern = req.internship
214     req.student.university = req.university
215 }
216
217 pred ApplicationCompleted(app: InternshipApplication) {
218     // Checks if an application is evaluated
219     app.result != PENDING
220 }
221
222 pred UniqueCredentials() {
223     // Ensures unique username and password for all users
224     all disj u1, u2: User | u1.username != u2.username
225         and u1.password != u2.password
226 }
227
228 pred HasCompletedInternship(s: Student) {
229     // Checks if a student completed an internship
230     some i: s.ongoingIntern | i.state = CLOSED
231 }

```

```

226 pred IsValidApplication(a: InternshipApplication) {
227     // Verifies an application is valid
228     a.result = PENDING and
229     a.application.state = OPEN
230 }
231 pred CheckStudentInternships(s: Student) {
232     // Ensures internships of a student have valid states
233     all i: s.ongoingIntern |
234         i.state in Status - OPEN
235 }
236 pred ValidateInternshipDates(i: Internship) {
237     // Validates that internship dates are consistent
238     i.startDate < i.endDate
239 }
240 // + - - - - - - - - - - - - - - - - - +
241 // |           TEMPORAL PREDICATES           |
242 // + - - - - - - - - - - - - - - - - - +
243
244 pred StudentSubscribers[s: Student]{
245     //pre-condition
246     s not in SandC.students
247     s.university in SandC.universities
248     //post-condition
249     always(s in SandC.students' and
250         SandC.students'=SandC.students+s)
251 }
252 pred CompanySubscribers[c: Company]{
253     //pre-condition
254     c not in SandC.companies
255     //post-condition
256     always(c in SandC.companies' and
257         SandC.companies'=SandC.companies+c)
258 }
259 pred UniveristyContract[u: University]{
260     //pre-condition
261     u not in SandC.universities
262     //post-condition
263     always(u in SandC.universities' and
264         SandC.universities'=SandC.universities+u)
265     //postpost-condition (only for fun)
266     some s: Student| (s.university'=u and s not in SandC.
267         students'
268         and s.ongoingIntern'=none and s.interApp'=none and

```

```

        always(
268     s in SandC.students'' and
269     SandC.students''=SandC.students+s))
270 }
271 pred LoadCV[s: Student, cvs: CV]{
272     //pre-condition
273     s in SandC.students
274     s.university in SandC.universities
275     s.interApp=None
276     s.cv =None
277     cvs.~cv=None
278     //post-condition
279     all s1: Student-s| s1.cv=s1.cv'
280     s.cv'=cvs and cvs.~cv'=s
281     s in SandC.students'
282     s.university in SandC.universities'
283     SandC.students'= SandC.students
284 }
285 pred InterApp[s: Student, ia: InternshipApplication]{
286     //pre-condition
287     s.interApp!=None and not ia in s.interApp
288     s in SandC.students
289     //post-condition
290     all s1: Student-s| s1.interApp=s1.interApp'
291         ia in s.interApp'
292         s.interApp'=s.interApp+ia
293 }

294
295 // + - - - - - - - - - - - - - - - +
296 // |                               |
297 // + - - - - - - - - - - - - - - - +
298
299 assert EachStudentHasUniversity {
300     // Each student must be associated with exactly one
301     // university
302     all s: Student | one s.university
303 }
304
305 assert ValidInternshipState {
306     // Ensures all internships have valid states
307     all i: Internship | i.state in Status
308 }
309
310 assert AllApplicationsEvaluated {

```

```

308     // Verifies that all applications have been evaluated
309     all app: InternshipApplication | ApplicationCompleted
      [app]
310 }
311 assert ApplicationsToOpenInternships {
312     // Ensures applications are only made to open
      internships
313     all a: InternshipApplication |
314         a.application.state = OPEN
315 }
316 assert NoOverlapBetweenOpenAndOngoing {
317     // Checks there is no overlap between OPEN and
      ONGOING internships
318     no i: Internship | i.state = OPEN and i.state =
      ONGOING
319 }
320 assert ValidSupportRequest {
321     // Validates that all support requests refer to valid
      internships
322     all sr: SupportRequest | sr.internship in Internship
323 }
324 assert FeedbackHasUser {
325     // Ensures every feedback is linked to a user
326     all f: Feedback | some u: User | f in u.feedback
327 }
328 assert InternshipOngoingValidity {
329     // Validates that ONGOING internships have consistent
      dates
330     all i: Internship |
331         i.state = ONGOING implies i.startDate < i.endDate
332 }
333
334 // + - - - - - - - - - - - - - - - - - +
335 // |                               RUN                               |
336 // + - - - - - - - - - - - - - - - - - +
337
338 run {
339     some s: Student, c: Company |
340         s in SandC.students and c in SandC.companies
341 }

```

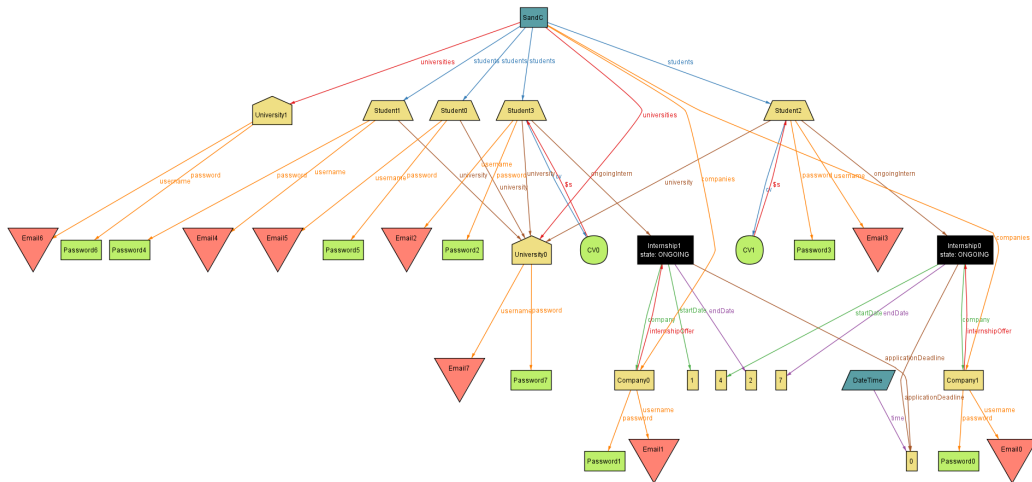


Figure 7: [run](#) General

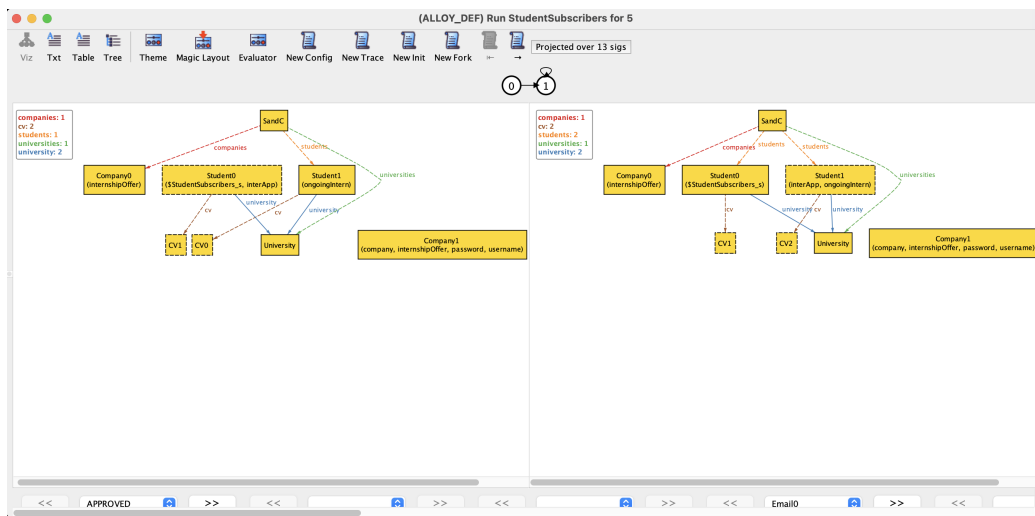


Figure 8: [run](#) StudentSubscribers

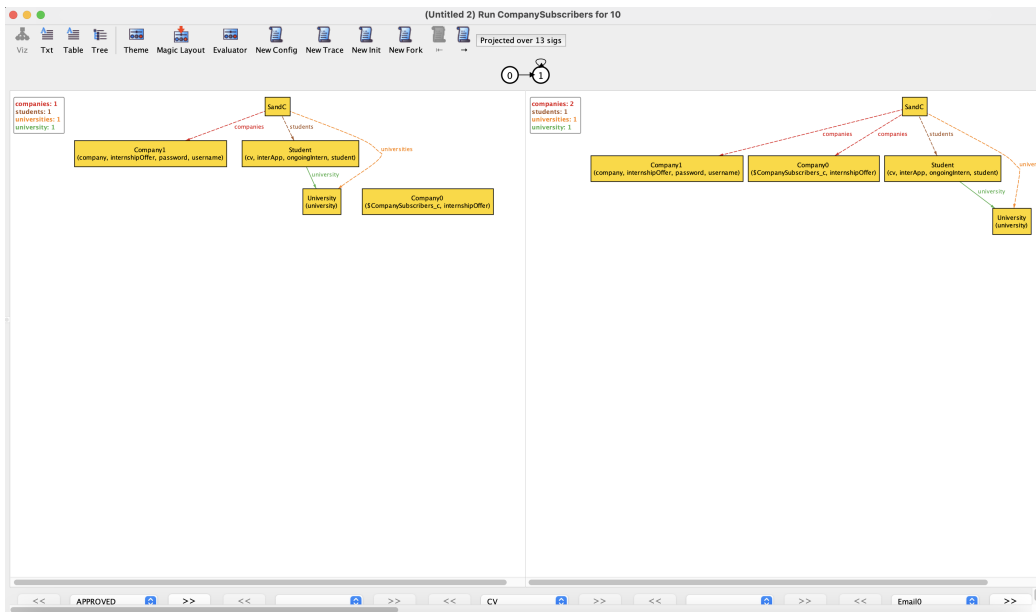


Figure 9: [run](#) CompanySubscribers

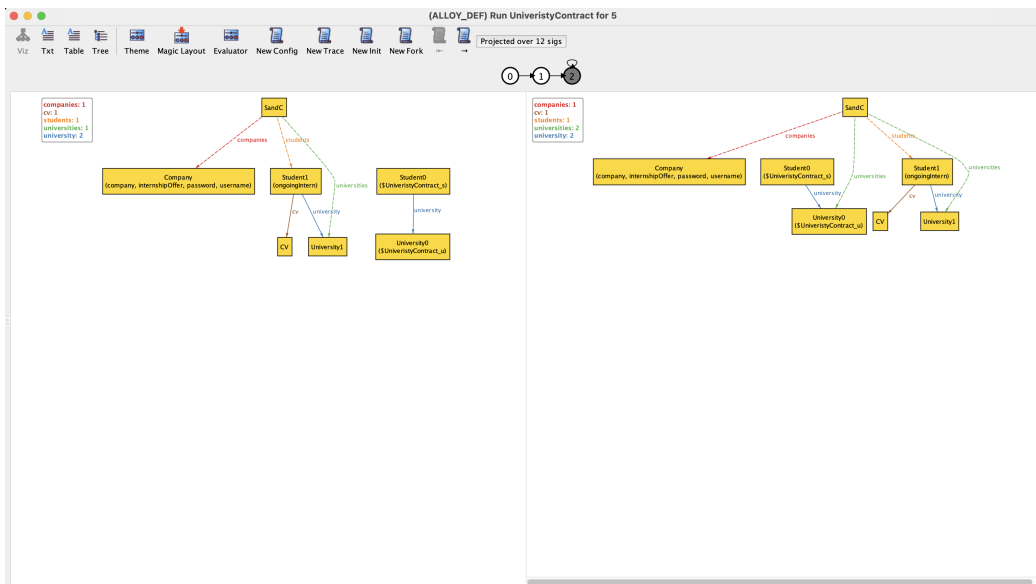


Figure 10: [run](#) UniveristyContract+University

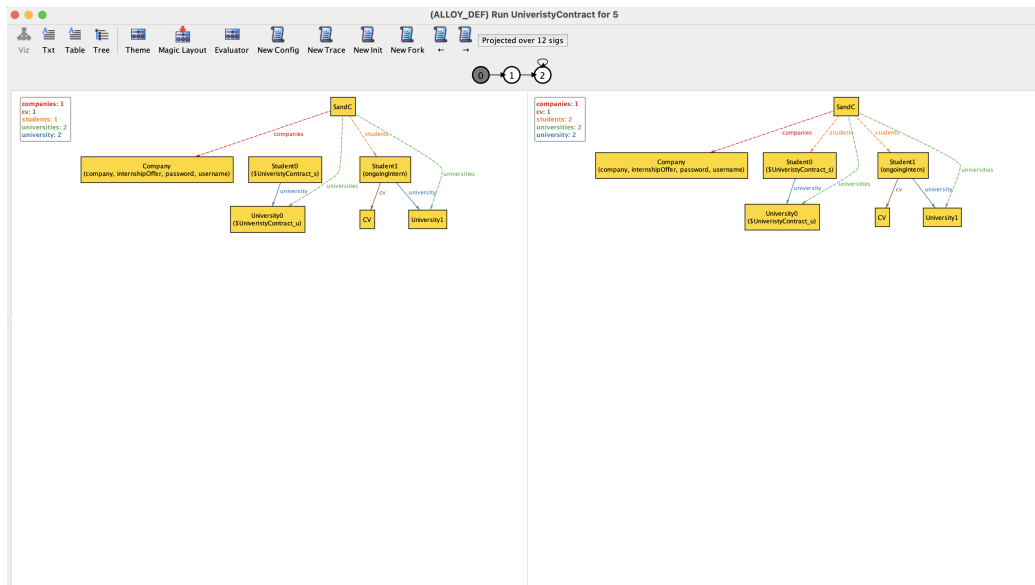


Figure 11: [run](#) UniveristyContract+Student Subscribes

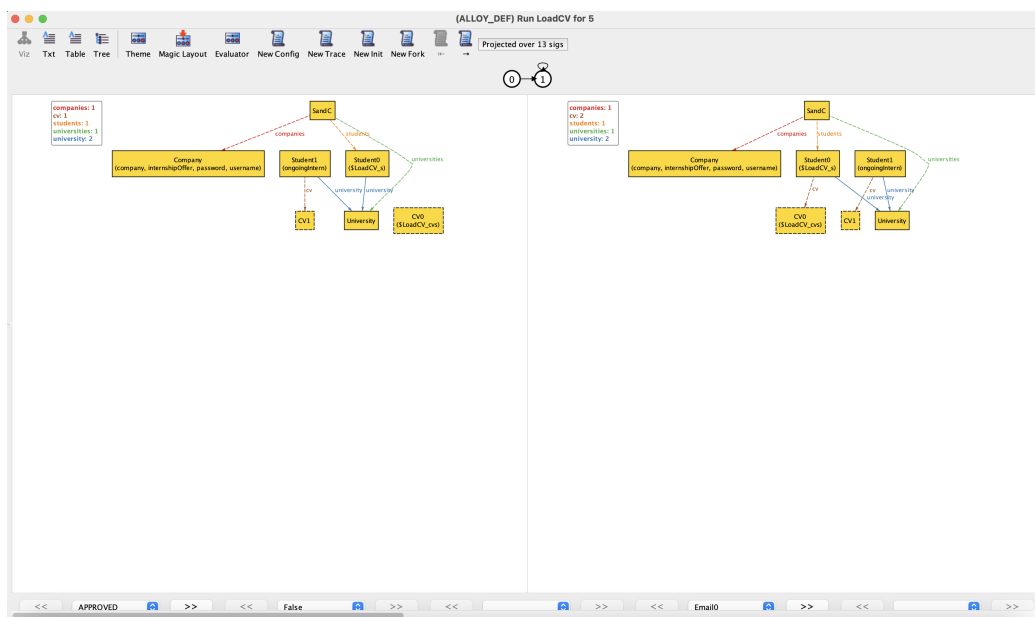


Figure 12: [run](#) LoadCV

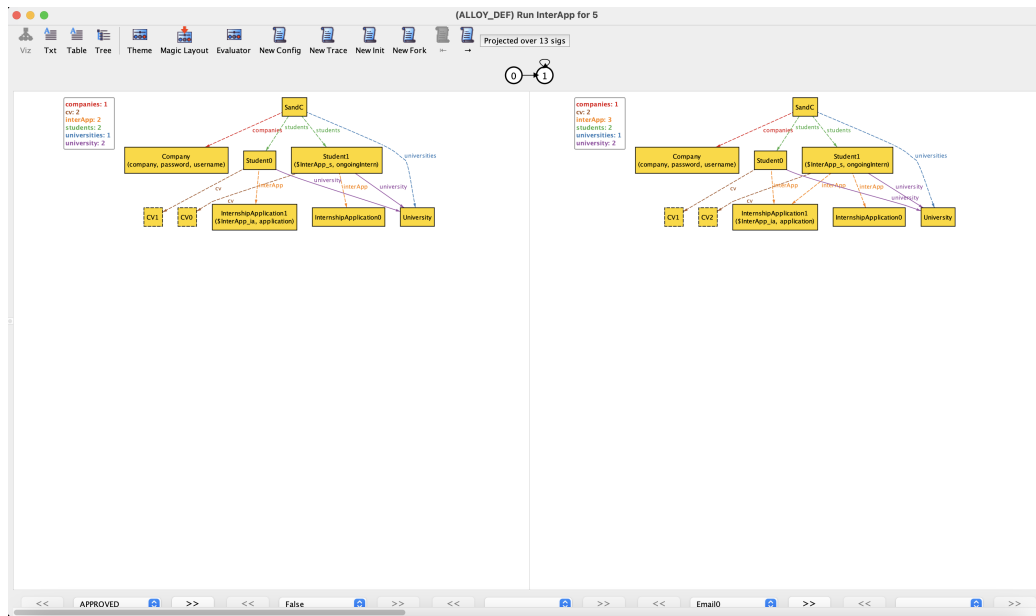


Figure 13: [run InterApp](#)

5 Effort Spent

Nardi Andrea

Activity	Hours Spent
Introduction	9
Overall Description	18
Specific Requirement	18
Formal Analysis Using Alloy	4

Pesaturo Christian Giovanni

Activity	Hours Spent
Introduction	10
Overall Description	14
Specific Requirement	20
Formal Analysis Using Alloy	5

Pinessi Andrea

Activity	Hours Spent
Introduction	9
Overall Description	15
Specific Requirement	19
Formal Analysis Using Alloy	6